Velocity Vista: Conceptual Design of a Novel Hydrogen-Powered Short-Range Aircraft

A Project Report

Submitted by

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BACHELOR OF ENGINEERING

in

AEROSPACE ENGINEERING

Under the Guidance

Of

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UNIVERSITY OF ENGINEERING(UIE)

CHANDIGARH UNIVERSITY

BONAFIDE CERTIFICATE

Certified that this project report "**Velocity Vista: Conceptual Design of a Novel Hydrogen-Powered Short-Range Aircraft**" is the BONAFIDE work of

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Who carried out the work under my supervision certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation based on which a degree or award was conferred on an early occasion on this or any other candidate.

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The report of the project work submitted in partial fulfilment for the award of Bachelor of Engineering in Aerospace Engineering of Chandigarh University were evaluated and confirmed to be the reports of the work done by the above students and then evaluated.

INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

This project though done by us would not have been possible, without the support of various people, who by their cooperation have helped us in bringing out this project successfully.

We thank our Professors & Head Dr. Dharmahinder Chand who had always served as an inspiration for us to perform well.

We would like to express our unbounded gratefulness to **Ms. Moniya Pal** for her valuable guidance and encouragement on the project.

At last, we would like to thank all the faculty members and supporting staff and the seniors for the help they extended to us for the completion of this project.

Finally, we must state that no height is ever attained without some sacrifices being made at some point, and it is in this regard that we owe a special debt to our parents and friends for their kind love and care during the entire period of time.

ABSTRACT

This report presents a detailed design of a short-range, high-capacity aircraft aimed at addressing the issue of overcrowded airports in major economic hubs. The aircraft is specifically designed to offer a solution without the added size and cost associated with long-range capabilities. Key features of the design include a twin-engine layout positioned at the rear of the fuselage, complemented by a cruciform tail configuration to ensure structural stability and accommodate the GE B67B engine nacelles.

The aircraft is tailored to accommodate 230 passengers in a two-class configuration and has a maximum range of 3500 nautical miles at a cruising altitude of 35,000 feet and a cruising speed of Mach 0.8. With a maximum take-off weight of 160,000 pounds, the aircraft is optimized for efficient performance.

An innovative aspect of the design is the use of hydrogen as a fuel source, aimed at reducing the carbon footprint of air travel, revolutionizing the industry, and cutting down operational costs. With these capabilities, the aircraft is positioned to lead the commercial aviation market in terms of efficiency and play a significant role in reducing congestion at airports worldwide.

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Nomenclature

AC	Aerodynamic Center
$\dot{m}$	Mass Flow Rate
CG	Center of Gravity b Wingspan
CFD	Computational Fluid Dynamics
CD	Drag Coefficient
FAR	Federal Aviation Regulation
Cl	Lift Coefficient
MAC	Mean Aerodynamic Chord
CM	Pitching Moment Coefficient
RFP	Request for Proposal
D	Diameter
ULD	Unit Loading Device
Df	Fuselage Diameter
L	location
L_f	Fuselage Length
P	Power, Loading
T/WTO	Thrust-to-Weight Ratio
S	Wing Area
v	Velocity
W	Weight
WTO/S	Take-off Wing Loading
TO	Take Off

1. Introduction

In the realm of international commerce, the demand for efficient short-range travel between economic centers has intensified. The absence of a high-capacity, short-range transport jet has led to congestion issues at major airports worldwide. Delays, as highlighted by the Department of Transportation, constitute a significant portion of passenger complaints, amounting to about 40% of all grievances. The growth of emerging economic hubs such as India, UAE, Qatar, etc. exacerbates this congestion dilemma. This, coupled with dwindling fuel reserves and volatile oil prices, creates an unstable environment for both airlines and passengers, underscoring the critical need for efficient airspace utilization.

The challenge facing modern airliners stems from insufficient seating capacity to meet evolving consumer and environmental demands. Many medium- to short-haul aircraft were designed during a time when air travel demands were less pronounced. World Bank data reveals a staggering increase in international air travel demand from 310 million users annually in 1970 to 4.25 billion today. Projections indicate a continued 5.2% annual growth over the next two decades, leading to a surge in active fleet numbers and exacerbating congestion problems.

To address these challenges, our proposed aircraft design focuses on doubling passenger capacity and enhancing the propulsion system to address environmental and consumer concerns. The envisioned propulsion system features a high bypass ratio (10:1) engine capable of running on hydrogen fuel, offering cost savings and reduced greenhouse gas emissions. Comparing fuel costs, with current short-range aircraft like the Airbus A220 carrying a maximum of 230 passengers and a 5,970-gallon fuel tank, operational costs per flight amount to $34,570. Transitioning to hydrogen, priced at $14 per kg (equivalent to $5.60 per gallon of fuel), and with anticipated reductions to $8 per kg in the future, operational costs per flight drop to approximately $18,528, without carbon tax due to hydrogen's cleaner properties. This 46% reduction in operational costs not only boosts airlines' profitability but also curtails their contribution to greenhouse gas emissions, aligning with sustainability goals.

The high-capacity short-range transport jet outlined in this report will comply with CFR Title 14- Part 25 regulations, with a planned service entry date of 2029. Key technologies will be developed to ensure the feasibility of hydrogen usage, including advancements in fuel tank technology for pressurized storage, fire prevention systems, fuel dump mechanisms, and improvements in heat exchangers for safe and efficient heating of hydrogen fuel.

1.1 Project Outline

The report's introduction outlines the challenges faced by modern airliners, addressing these challenges, and the design decisions made to tackle them. Subsequent sections delve into a comprehensive trade-off analysis of major components and system-level design choices made by the team. A comparative analysis with existing competing aircraft is also included to ensure that the proposed design surpasses current offerings. An engine matching study is then conducted, aligning the rendered design to meet and exceed performance requirements under specified conditions. A weight analysis follows to determine optimal placement for wings and landing systems. The final configuration is meticulously analyzed and refined to minimize drag during cruise flight. Lastly, performance metrics such as takeoff, cruise, and landing are rigorously analyzed to ensure compliance with the efficient future specifications. The report concludes with key findings and recommendations for future development.

2. Concept Evolution & Comparative Study

2.1 Design Goal

The primary objective of this semester project is to design an aircraft with a seating capacity similar to the Airbus A220, optimized to accommodate 230 passengers. Additionally, the propulsion system will be optimized by utilizing Hydrogen Fuel instead of traditional Jet fuel, aiming to enhance sustainability and reduce carbon emissions. The aircraft design will be tailored to a reference mission similar to that of the Airbus A220, focusing on typical distances between major economic hubs.

The main design goals for this aircraft include achieving cost efficiency while either maintaining or improving reliability and repairability compared to existing aircraft. The mission specifications for the designed aircraft are summarized in Table 1 for quick reference.

Table 1: Design Specifications

Velocity Vista	
Min No. Passengers	230
Cruise Speed	890 Km/hr
Cruise Altitude	35,000 ft
Range	3500 (n.mi)
Maximum Take-off length	1.5341 Km

The expected mission profile for the aircraft encompasses various phases such as takeoff, climb, cruise, descent, and landing, with specific performance parameters tailored to meet the demands of the reference mission and optimize overall operational efficiency.

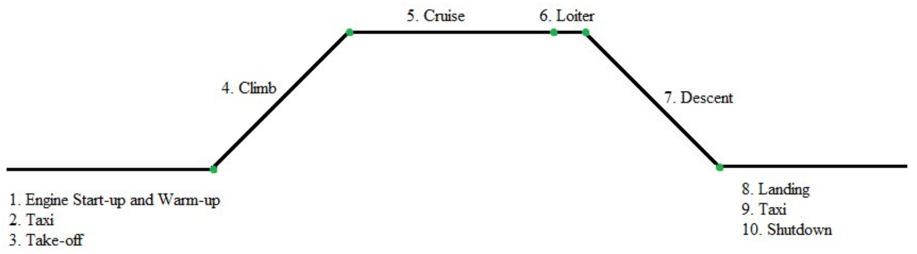


Figure 1: Mission Profile for Velocity Vista

2.2 Trade Study

In the realm of aircraft design, making informed decisions is paramount to creating an optimal and efficient aircraft that meets project objectives and stakeholder requirements. A trade study forms a crucial part of this decision-making process, offering a systematic approach to evaluating and comparing various design options, components, and systems.

The purpose of this trade study is to delve into the intricacies of aircraft design, considering factors such as performance, cost, weight, reliability, and environmental impact. By conducting a comprehensive analysis of different design elements, we aim to identify the most viable and effective solutions that align with the project's goals and constraints.

Through this trade study, we seek to answer critical questions about the aircraft's configuration, propulsion system, avionics, structural design, materials, and aerodynamics. By exploring trade-offs and examining the strengths and weaknesses of each option, we can make well-informed decisions that optimize the aircraft's overall performance, efficiency, and sustainability.

Airbus A220

The A220-100, powered by Pratt & Whitney PurePower PW1500G geared turbofan engines, is a powerful aircraft capable of operating in hot-and-high and city-centre airports. With a 3,450nm range and exceptional airfield performance, it is the largest aircraft capable of operating in London City Airport, providing a built-in advantage in challenging airport environments.

The A220-100 offers a configurable cabin with two flex zones, allowing operators to customize modular elements for an exceptional passenger experience. The A220 Family aircraft are known for their low noise levels, providing a quiet and comfortable environment during all flight phases. Wide seats and ample overhead storage create personal space without compromising headroom, and the cabin management system allows crews to control the aircraft's interior environment, including entertainment and mood lighting.

Airbus A310

Airbus' A310 is the most profitable 200-seat jetliner in its class, offering lower operating costs and better economy than other airlines. Since its 1983 service entry, the aircraft has provided maximum flexibility for operators, accommodating between 190 and 230 passengers in a typical three-class layout. With two engine options – General Electric's CF6-80C2 and Pratt & Whitney's PW4000 – the A310 offers reduced fuel-burn, lower noise levels, and CO2 emissions.

The A310 is a wide-body jetliner with a 222-inch fuselage cross-section, offering ample space and comfort for passengers. Its flexible seating arrangements and options for galley, lavatory, and attendant seats allow operators to customize the cabin to meet their needs. The A310 also features large overhead storage for optimal carry-on baggage capacity and draught-free air conditioning with temperature control in three areas. With modern in-flight entertainment and Airbus' high-quality design standards, the A310 ensures a relaxing and enjoyable flight experience for all passengers.

Airbus A320 Neo

The A320neo Family is firmly established as the most modern and best-selling single aisle fleet in the world**,** reaching new heights in efficiency, and achieving great distances of up to 4700nm with the A321XLR. With passenger seating ranging from 120 to 244**,** the versatility of the A320neo Family ensures that when market demands change, you will be ready to deliver the right capacity, at the right time, all for the right price.

Each aircraft in the A320neo Family is the same aircraft in a different size, so your pilots, crews, and mechanics only have to learn one in order to master them all. And with Airbus' signature Airspace cabins, your passengers will enjoy a superior travel experience while your business enjoys a better bottom-line.

Airbus A330

The A330neo, launched in 2014, offers lower operating costs by combining A350's enhanced wing technologies with Rolls-Royce Trent 7000 engines. It also features Airbus' Airspace™ cabin, making it the quietest and most comfortable in its category. The A330neo is a seamless and low-risk replacement for customers already flying A330 aircraft, as it retains commonality across 95% of its parts and 85% of its tooling by value. Crews can easily transition to the

A330neo after just a half-day of training, and pilots can fly both aircraft types under one license, requiring only 8 days of training.

The A330neo, a new Airbus aircraft, offers passengers ample personal space with wider seats and a comfortable Airspace cabin. It also reduces fuel burn CO2 emissions by double, demonstrating Airbus' commitment to sustainable air transportation. The A330neo also serves as a modern widebody freighter, corporate jet, and is deployed by military forces in its Multi Role Tanker Transport (A330 MRTT) variants.

Airbus A340

Airbus' A340s, part of its modern aircraft line, utilize digital fly-by-wire controls and a modern cockpit. They share similar airframe structures and components with the A330, providing a true aircraft family with different versions. This system facilitates worldwide support, maintenance, and access to qualified pilots, with Airbus fully supporting the A340.

The A340 is a four-engine commercial jetliner that offers superior operating economics and excellent passenger comfort. It can fly long and demanding routes, providing greater range at lower costs. Airbus offers four variants, including the A340-200, A340-300, A340-500, and A340-600, with extended operating ranges of up to 9,000 nautical miles. The A340 has flown over 600 million passengers and has been ordered over 370 times.

Airbus A350

The A350 Family, consisting of two versions: the A350-900 and the longer fuselage A350-1000, offers efficient flight capabilities on short-haul to ultra-long-haul routes up to 9,700nm. With advanced cabin technology, high air quality, and integrated connectivity, passengers can arrive at their destinations feeling refreshed and relaxed. Smart galleys and comfortable crew rest areas provide a pleasant working environment, ensuring the well-being of passengers.

Airbus A380

The A380 has set a new standard for the global aviation industry. Not only did it usher in a new era for passenger comfort, the A380 also raised the bar for environmental standards with its low fuel consumption per passenger and low noise levels – as well as reduced CO2 and NOx emissions, which has been passed on to future aircraft generations. The A380, the largest airliner ever built, offers pilots familiarity with its sidestick control, similar to the A320, A330/A340, or A350 Families aircraft. The A380's Brake-to-Vacate system helps reduce airport congestion and runway time by allowing pilots to select a runway exit during landing. This system uses auto-flight, flight controls, and auto-brake systems to regulate deceleration after touchdown, ensuring the aircraft reaches a specified exit at the correct speed and avoiding runway excursions.

Boeing 737 Max

The 737 MAX delivers enhanced efficiency, improved environmental performance and increased passenger comfort to the single-aisle market. Incorporating advanced technology winglets and efficient engines, the 737 MAX offers excellent economics, reducing fuel use and emissions by 20 percent while producing a 50 percent smaller noise footprint than the airplanes it replaces. Additionally, 737 MAX offers up to 14 percent lower airframe maintenance costs than the competition. Passengers will enjoy the Boeing Sky Interior, highlighted by modern sculpted sidewalls and window reveals, LED lighting that enhances the sense of spaciousness and larger pivoting overhead storage bins.

Embraer E175

The Embraer E-Jet family is a series of four-abreast, narrow-body, short- to medium-range, twin-engined jet airliners designed by Brazilian aerospace manufacturer Embraer. The E-Jet was created as a complement to the ERJ family, Embraer's first jet-powered regional jet. It was larger than any prior aircraft built by the company and was introduced in 1997. The first prototype E-Jet conducted its maiden flight in 2002, and production began in 2004. The E170 was delivered to LOT Polish Airlines in 2004, and later, the E190 and E195 were introduced. The E-Jet family has been a commercial success, serving lower-demand routes and offering amenities and features of larger jets. It is commonly used by mainline and regional airlines worldwide, particularly in the United States.

The Embraer 175 is a slightly stretched version of the E170, first entering revenue service with Air Canada in 2005. It seats around 78 passengers in single-class, 76 in dual-class, and up to 88 in high-density configurations. It competes with the Bombardier CRJ900 in the market segment previously occupied by BAe 146 and Fokker 70. The E175 has been equipped with wider and more angled winglets since 2014, improving efficiency and consuming 6.4% less fuel than the original E175 aircraft. In 2017, Embraer announced the E175SC, limited to 70 seats, to replace the older 70-seat Bombardier CRJ700 with better efficiency and larger first class.

Sukhoi SuperJet 100

The Superjet 100, a 95-seat aircraft, was launched in 2007 at the KnAAPO assembly plant in Siberia. Its first flight took place in May 2008, followed by three more flights in December and July 2009. The aircraft achieved initial Russian certification in April 2009 and official certification from the Russian Interstate Aviation Committee Aviation Register in February 2011. The Superjet 100-75 and -95 variants can seat up to 78 and 98 passengers and are available in standard and long-range versions. The aircraft has a maximum cruise speed of Mach 0.81 and a range of 3,279km and 4,620km in the Superjet 100-95LR version.

2.3 Comparison

This section of the Aircraft Design Project Report delves into a comparative analysis of critical parameters across different aircraft models. By examining performance metrics, structural features, propulsion systems, avionics, materials, and environmental considerations, we aim to identify strengths, weaknesses, and areas for improvement.

Through this comparison, we seek to gain insights into industry standards, technological advancements, and best practices. This knowledge will guide strategic decisions to enhance our design's competitiveness, efficiency, safety, and sustainability.

Table 2: Comparative Data on weight of aircrafts

Aircraft Name	No. of Passengers	Payload Capacity (kg)	Fuel Weight (Kg)	Empty Weight (kg)	Range (km)	Cruise Speed (Km/h)
Airbus A220	220	39,000	79,000	80,200	9,600	850
Airbus A310	114	21,400	22,000	30,800	6,204	829
Airbus A320 Neo	180	29,000	26,730	42,600	6,380	828
Airbus A330 Neo	350	63,500	1,39,090	1,24,000	15,094	871
Airbus A340	320.5	66,000	1,95,000	1,29,000	16,670	871
Airbus A350	740	68,000	1,65,000	1,15,000	18,000	903
Airbus A380	689	84,000	3,10,000	2,76,800	15,700	903
Boeing 737 Max	184	21,800	37,648	41,413	7,130	839
Embraer E175	91	10,700	9,335	21,810	4,074	829
Sukhoi Superjet 100	118	13,790	15,805	24,880	4,578	829

Table 3:Comparative Data on Performance of aircrafts

Aircraft Name	Maximum Speed (Km/h)	Taper-Ratio	Take-Off Distance (m)	Lift-Drag Ratio	Fuselage Diameter (m)	Thrust-Weight Ratio
Airbus A220	945	0.28	2,300	18.5	5.64	0.59
Airbus A310	871	0.41	1,363	19.5	3.28	0.69-0.67
Airbus A320 Neo	871	0.28	1,500	17.5	3.95	0.57
Airbus A330 Neo	913	0.3	2,220	19.5	5.64	0.51-0.47
Airbus A340	913	0.28	3,100	18.5	5.64	0.43-0.32
Airbus A350	945	0.41	2,600	21.5	5.96	0.68-0.6
Airbus A380	945	0.3	2,900	19.5	7.14	0.45-0.35
Boeing 737 Max	876	0.2	1,300	16.5	3.76	0.57-0.6
Embraer E175	871	0.3	1,300	17.5	2.74	0.38-0.42
Sukhoi Superjet 100	871	0.3	1,731	17.5	3.24	0.62-0.61

Table 4: Comparative Data on Engine of aircrafts

Aircraft Name	Powerplant	Engine Weight (Kg)	Engine Thrust (Kn)	MTOW (Kg)	MLOW (kg)
Airbus A220	Pratt & Whitney PW1500G	2100	104.5	63,100	58,500
Airbus A310	General Electric CF6-80C2 or Pratt & Whitney PW4000	4100	236.3	1,64,000	1,20,000
Airbus A320 Neo	CFM International LEAP-1A or Pratt & Whitney PW1100G	2600	133.5	79,000	66,000
Airbus A330 Neo	Rolls-Royce Trent 7000	6500	320	2,51,000	1,91,000
Airbus A340	CFM International CFM56-5C	4300	194	2,75,000	1,70,000
Airbus A350	Rolls-Royce Trent XWB	7000	403	2,80,000	2,07,000
Airbus A380	Engine Alliance GP7000 or Rolls-Royce Trent 900	6300	342.5	5,75,000	3,94,000
Boeing 737 Max	CFM International LEAP-1B	2800	129.5	82,190	66,360
Embraer E175	General Electric CF34-8E	1600	182.3	38,600	34,500
Sukhoi Superjet 100	PowerJet SaM146 turbofan	2100	198	45,880	41,000

Table 5: Comparative Data on wings of aircrafts

Aircraft Name	Wing Area (m2)	Wing-Loading (Kg/m2)	Aspect Ratio	Wing Span (m)	Length (m)	Height (m)
Airbus A220	112.3	551.8	9.5	35.1	38.7	11.5
Airbus A310	219	427.4	9.1	43.9	46.7	15.8
Airbus A320 Neo	122.6	725.9	10.1	35.8	37.6	11.8
Airbus A330 Neo	362	605.8	9.7	64	63.7	16.8
Airbus A340	437	533.8	9.2	60.3	59.4	16.8
Airbus A350	443	666.7	9.5	64.8	66.9	17.1
Airbus A380	845	562.9	7.5	79.8	72.7	24.1
Boeing 737 Max	127	639.7	9.4	35.9	39.5	12.3
Embraer E175	92.1	580.4	8.4	26	31.7	10
Sukhoi Superjet 100	77.87	568.5	8.8	27.8	29.9	10.3

2.4 Performance Comparison

The comparison between the aircraft is done to see which aircraft is having the best performance and the efficiency. From this we can take the reference for our newly design aircraft to improve the maximum performance of the aircraft.

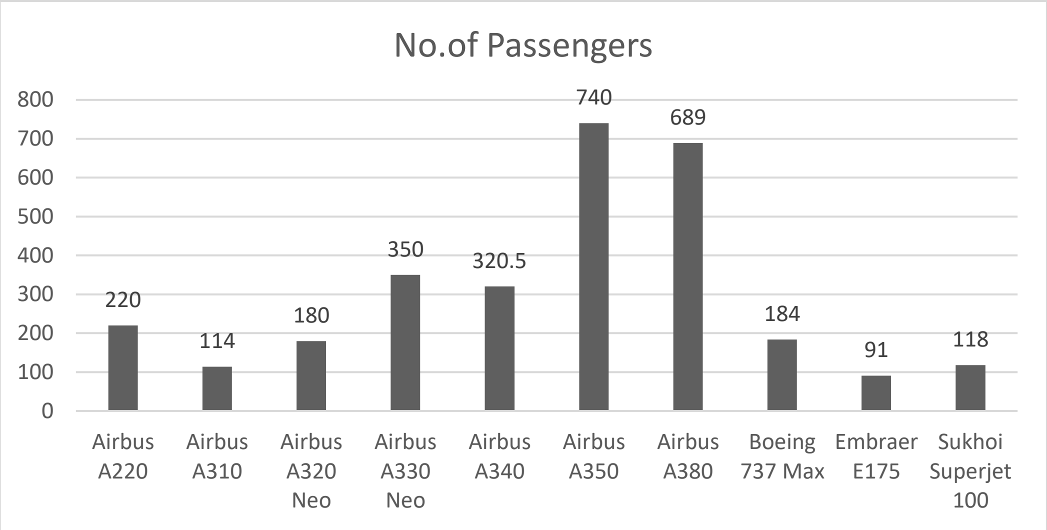


Figure 2: Comparative Graph of No. of Passengers

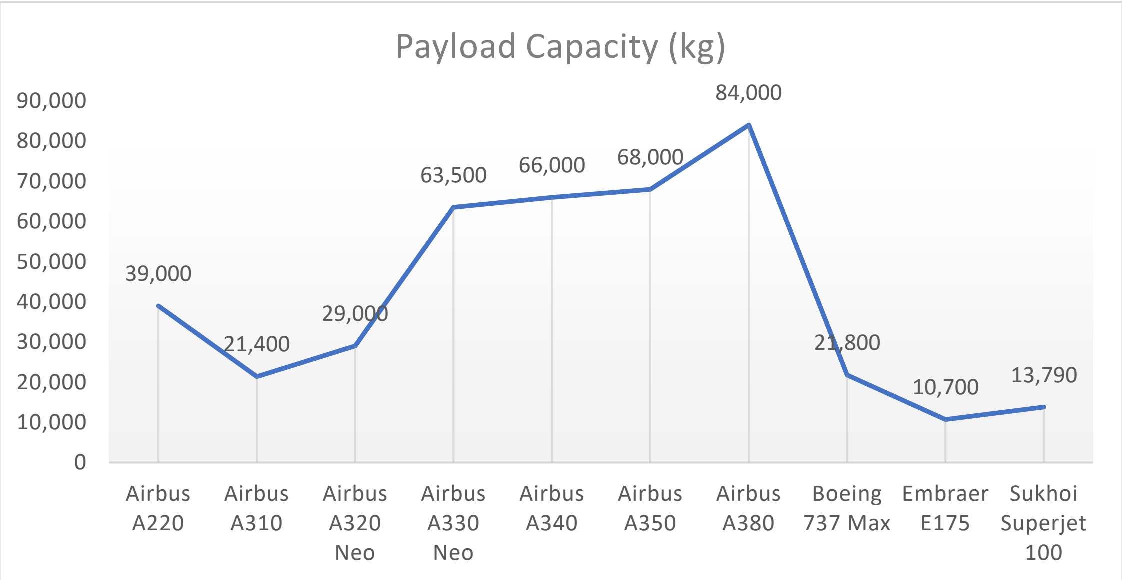


Figure 3: Comparative Graph of Payload Capacity

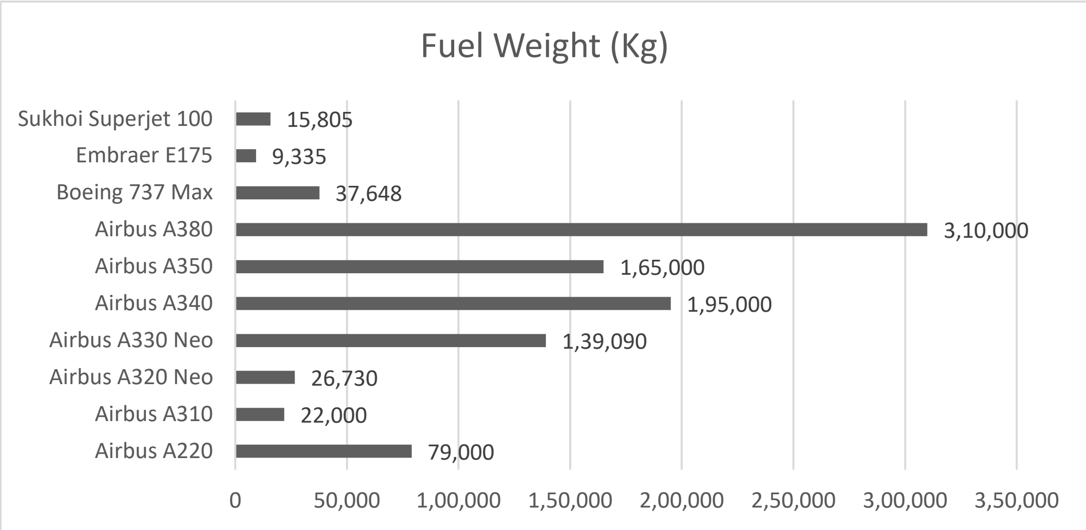


Figure 4: Comparative Graph of Fuel Weight

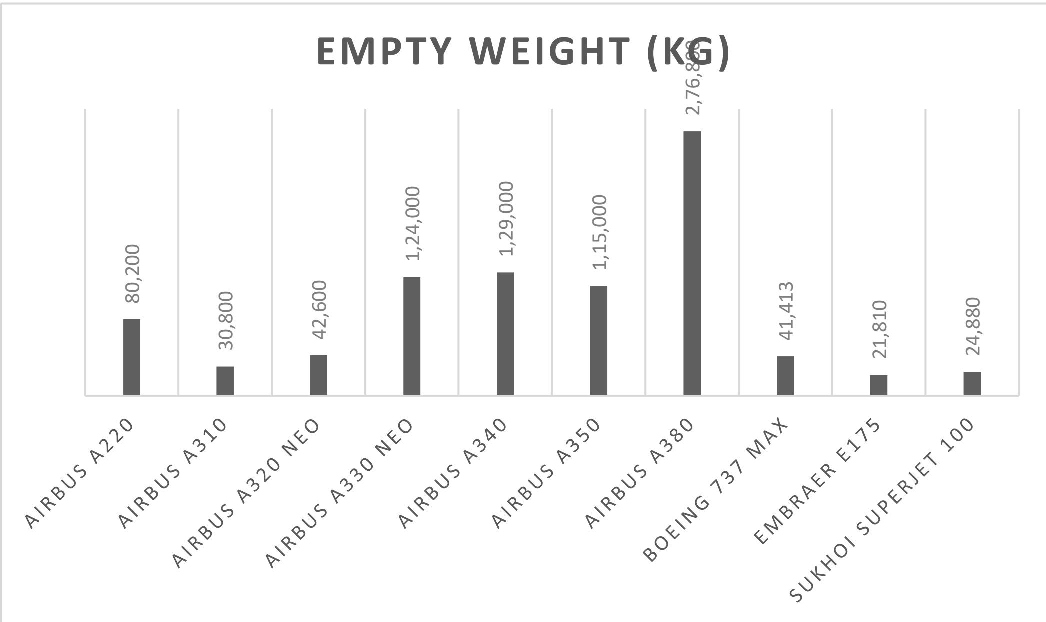
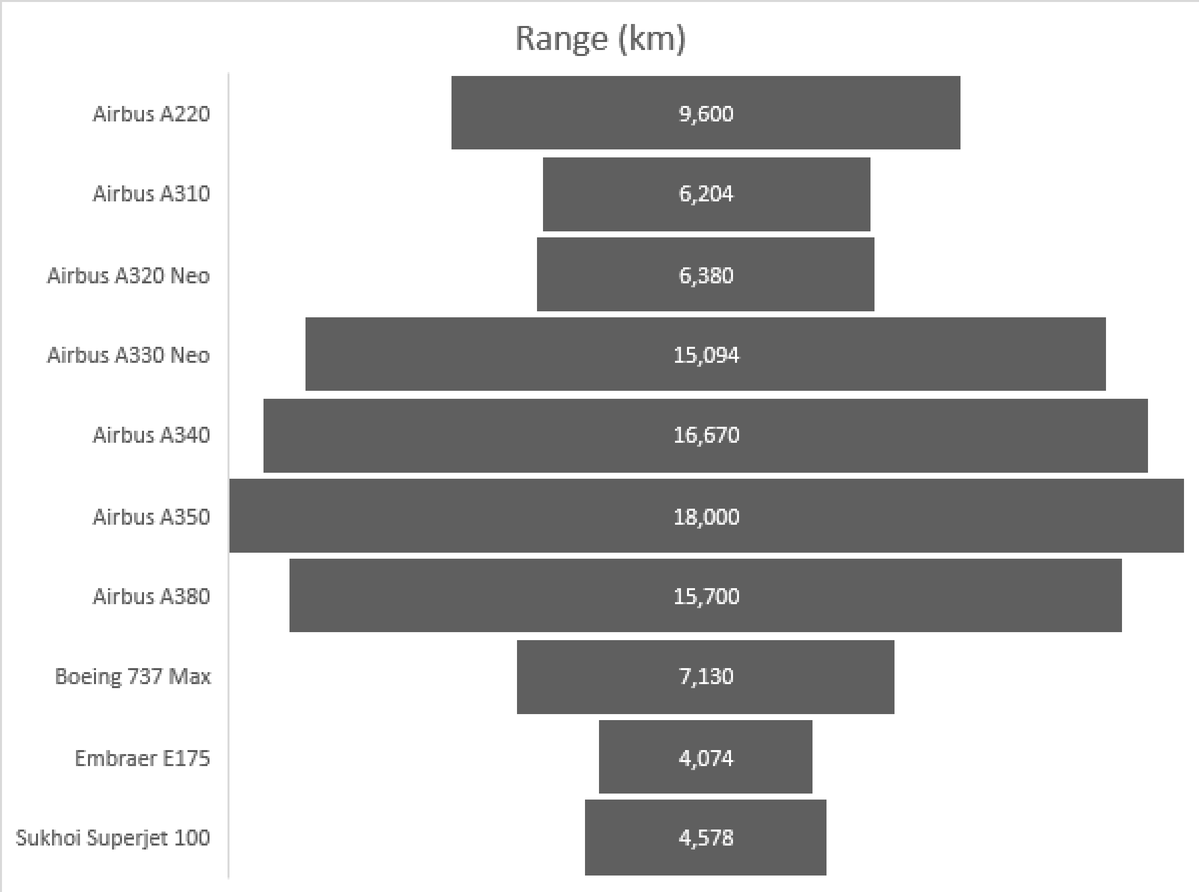


Figure 5: Comparative Graph of Empty Weight



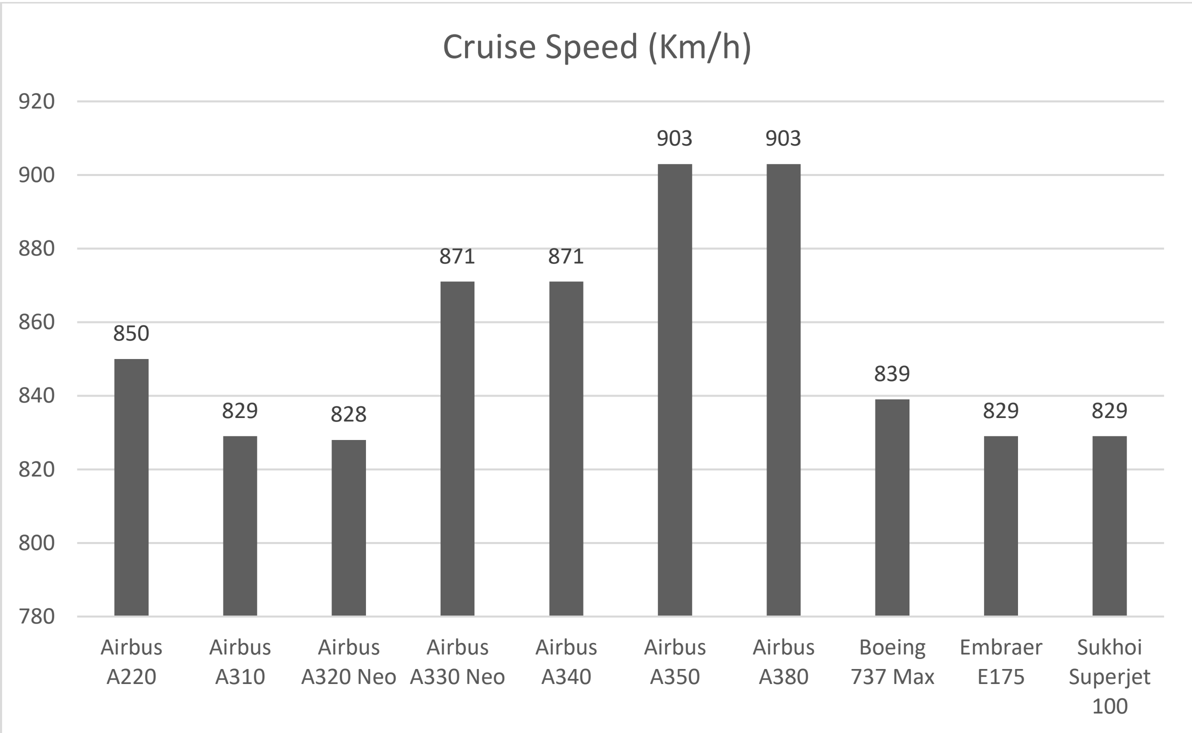


Figure 6: Comparative Graph of Cruise Speed

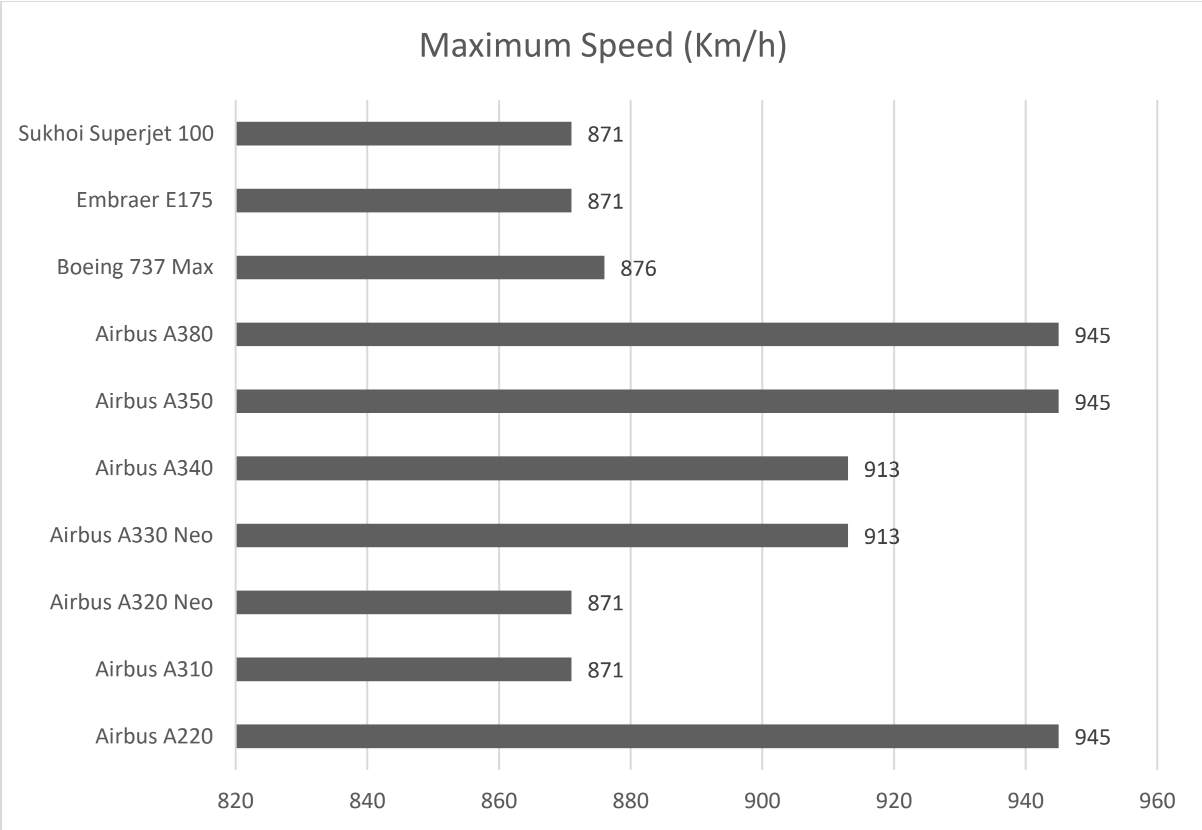


Figure 7: Comparative Graph of Maximum Speed

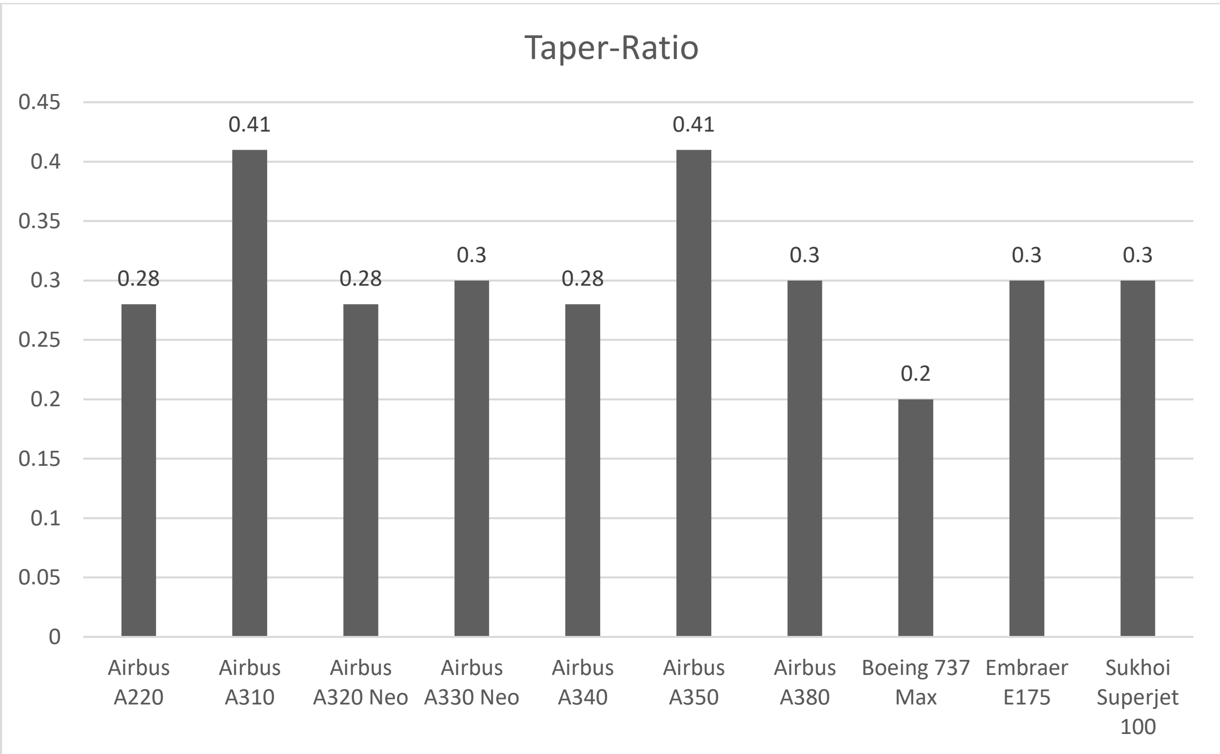


Figure 8: Comparative Graph of Taper Ratio

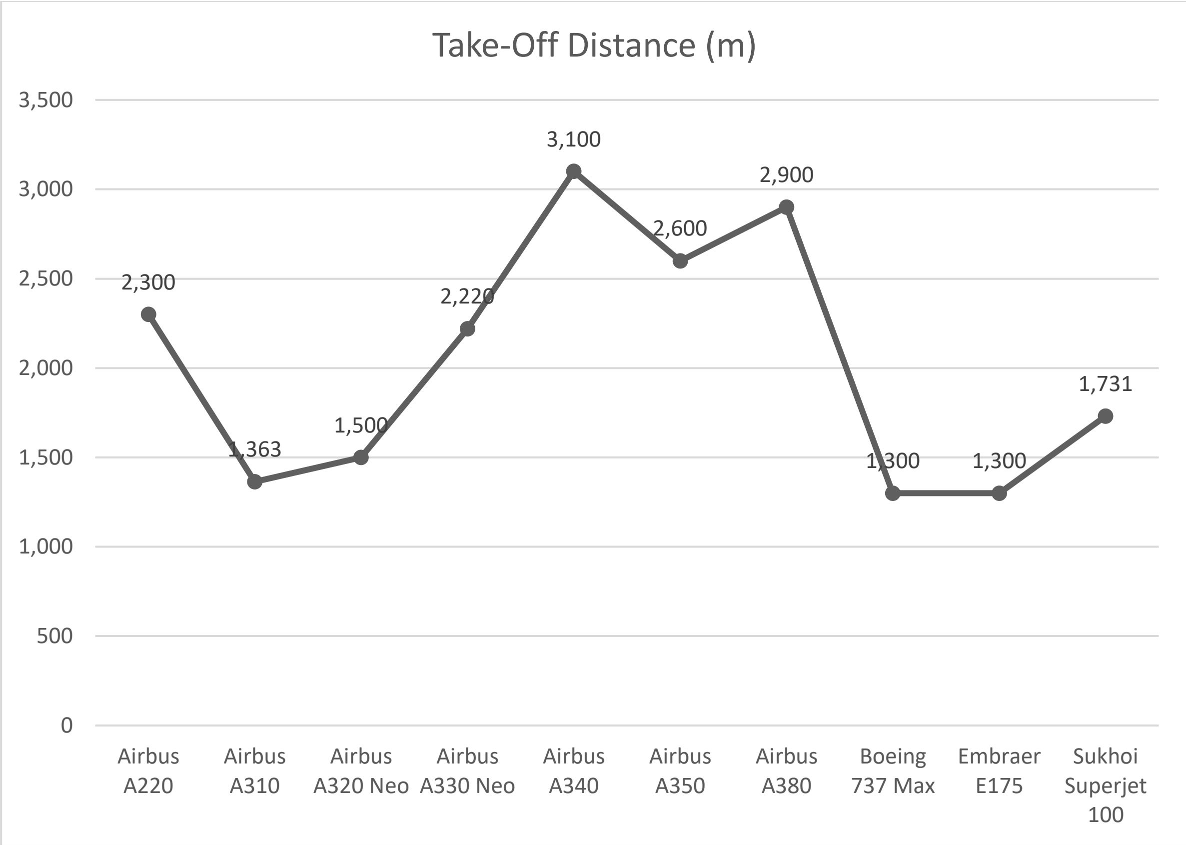


Figure 9: Comparative Graph of Take-Off Distance

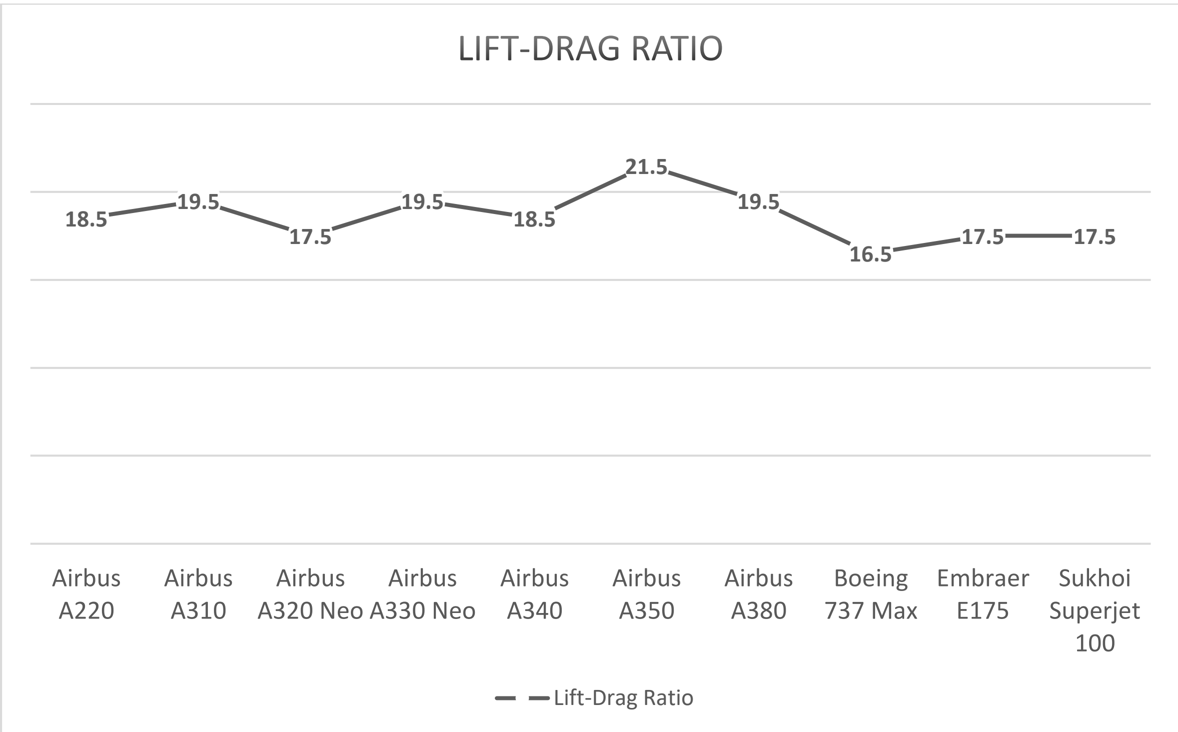


Figure 10: Comparative Graph of Lift-Drag Ratio

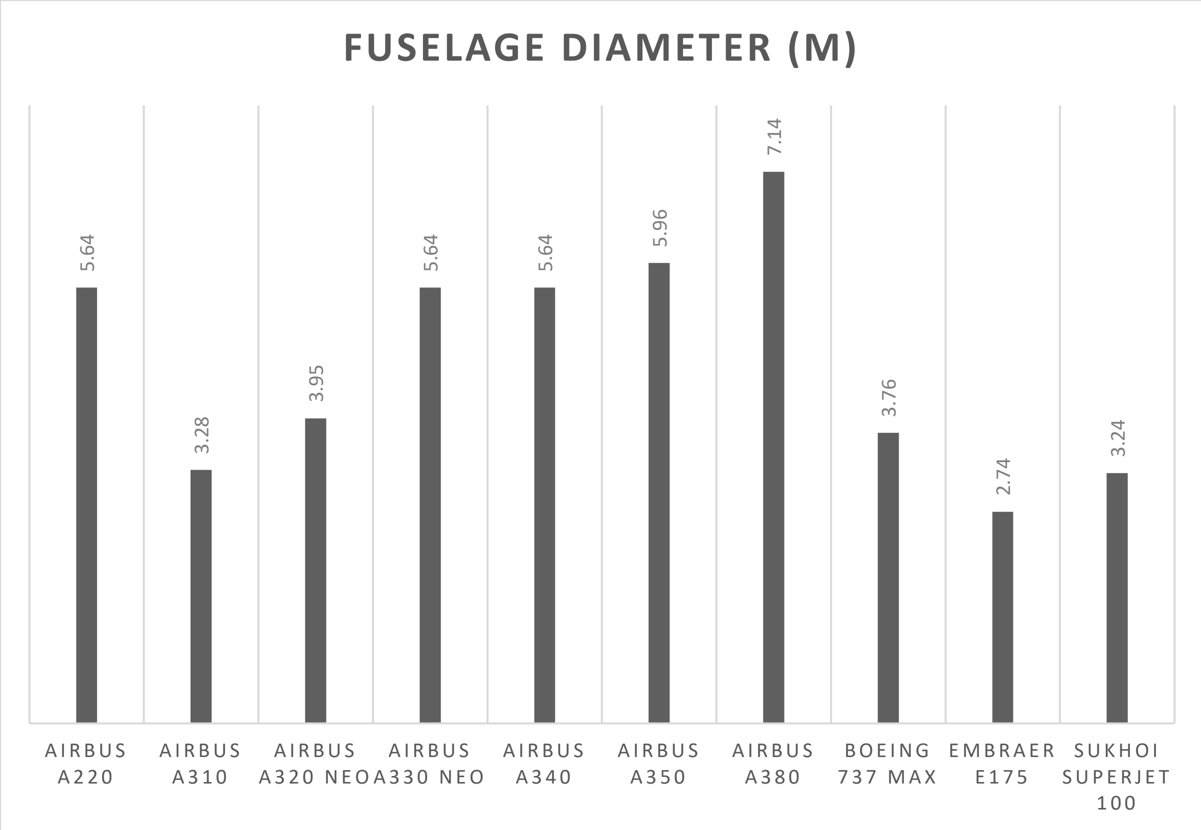
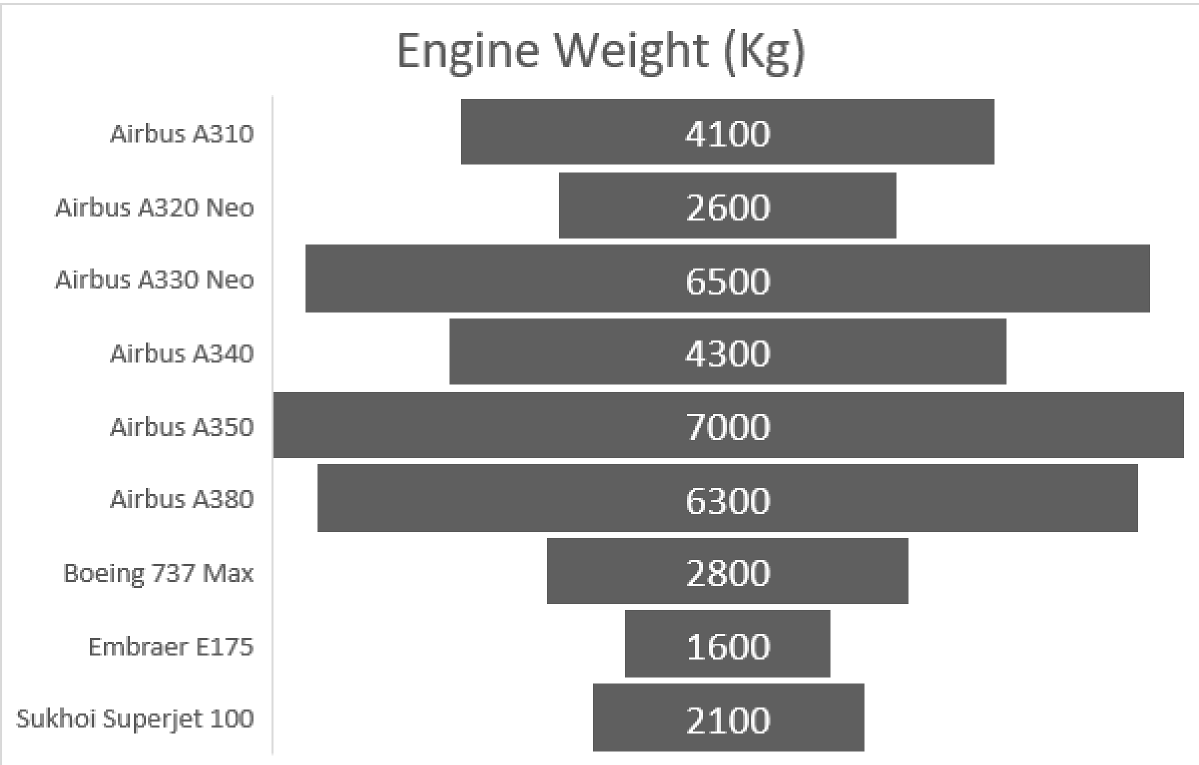


Figure 11: Comparative Graph of Fuselage Diameter



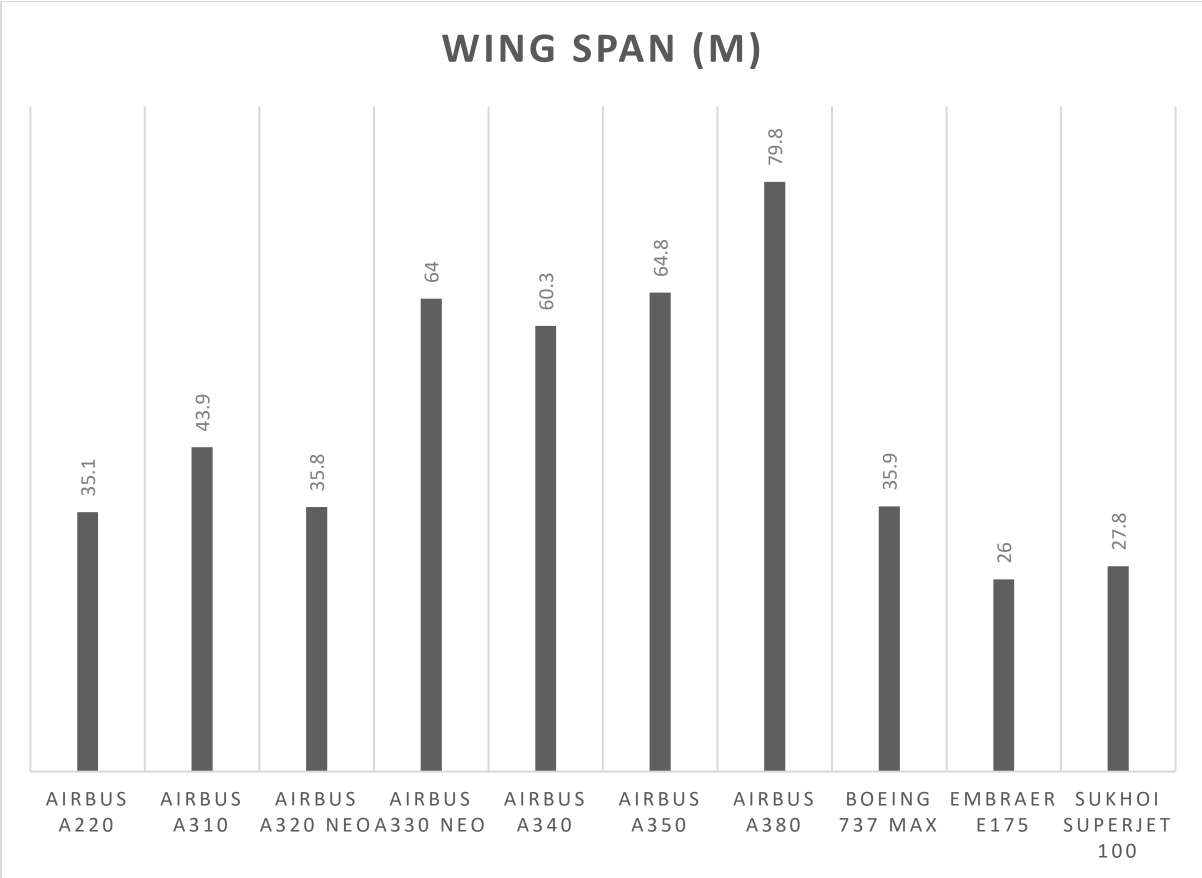


Figure 12: Comparative Graph of Wingspan

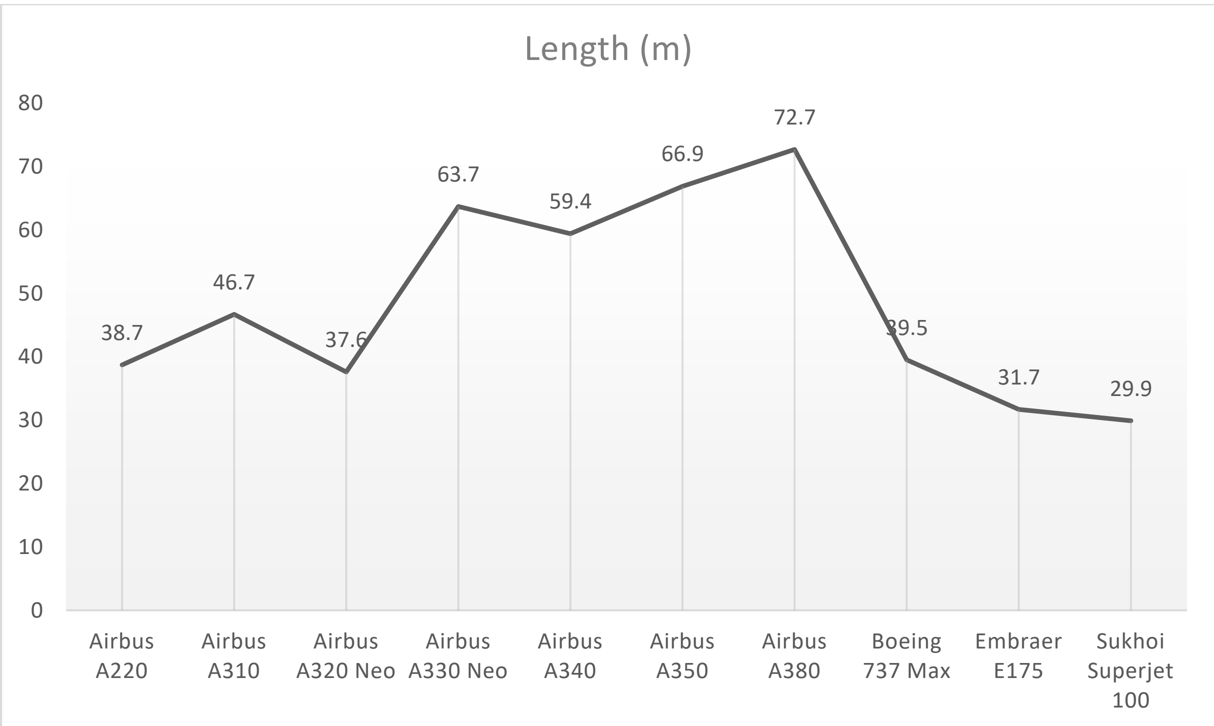
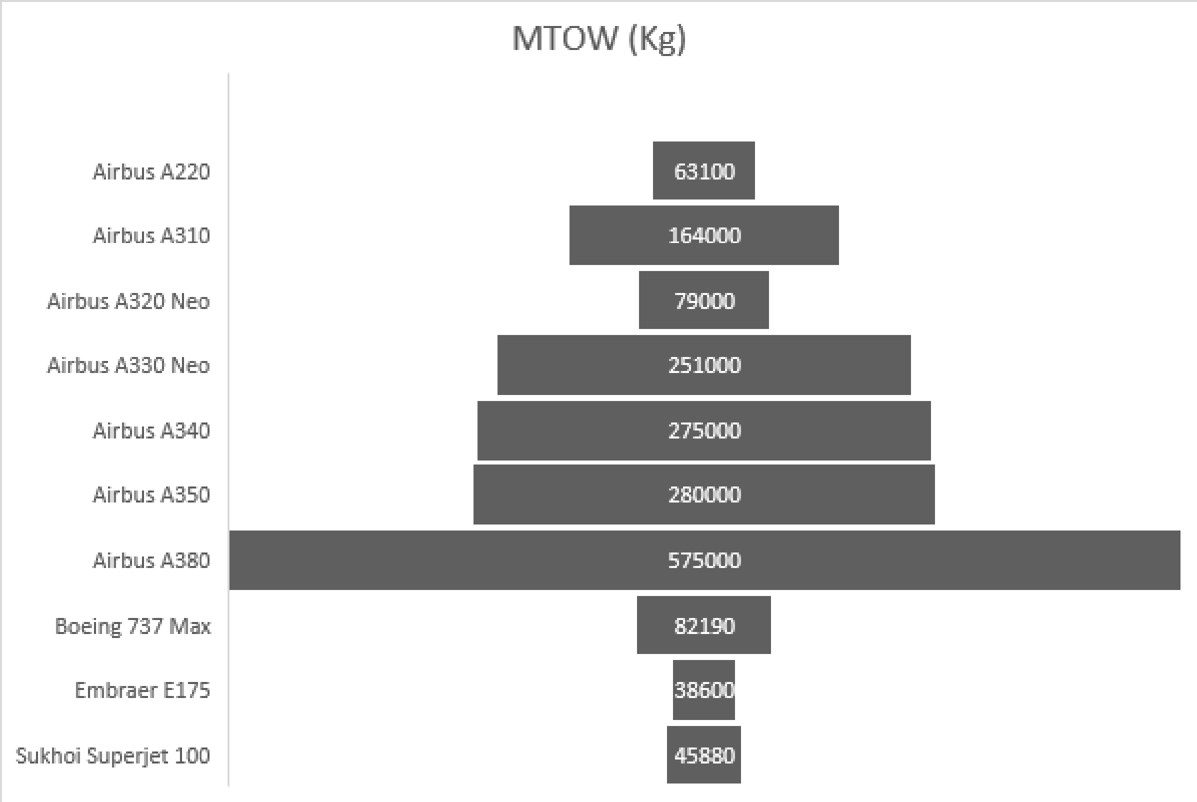


Figure 13: Comparative Graph of Length



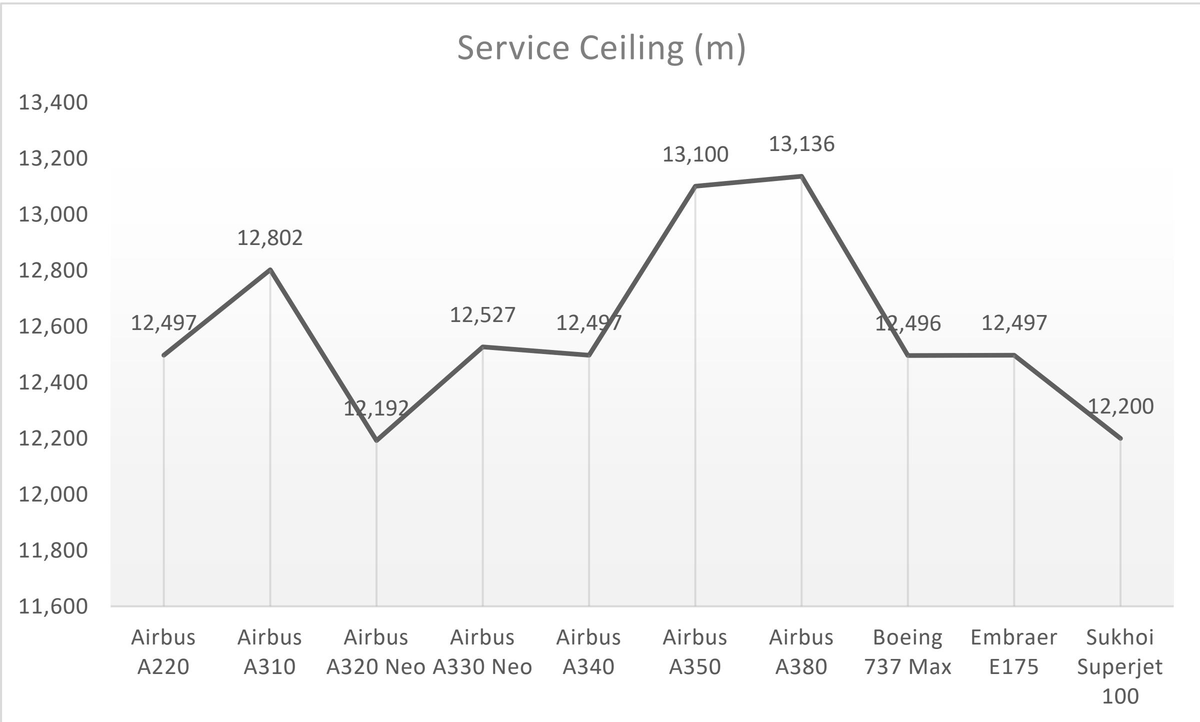


Figure 14: Comparative Graph of Service Ceiling

3. Preliminary Sizing

The take-off weight of the aircraft comprises the total weight of payload, fuel, and the operating empty weight of the aircraft. The payload weight includes the combined weight of passengers and baggage, calculated based on mission specifications assuming 200 lbs. for each of the 230 passengers.

Fuel weight is determined by estimating the fuel remaining after each flight phase to assess fuel consumption as a fraction of the take-off weight throughout the entire flight, including a one-hour loiter period.

3.1 **Weight Estimation**

The fuselage comprises the main body of the aircraft, housing the cockpit, cabin, and cargo compartments. It is typically constructed from lightweight materials such as aluminium alloy or composite materials to optimize strength and minimize weight.

The wings of the A220 are designed to provide lift and stability during flight. They are constructed using advanced aerodynamic principles and materials to maximize efficiency and structural integrity. The wings also house fuel tanks for storing the aircraft's fuel supply.

The landing gear includes the main landing gear (located under the wings) and the nose landing gear (located under the forward fuselage). These components support the aircraft during take-off, landing, and taxiing. They are engineered to withstand the forces encountered during these phases of flight and are retractable to reduce drag during cruising.

The A220 is typically equipped with two turbofan engines, such as the Pratt & Whitney PW1500G. These engines are mounted on pylons beneath the wings and provide the necessary thrust for propulsion. Engine weight is a significant factor in the aircraft's overall weight and balance calculations.

Fuel is stored in tanks located within the wings and fuselage of the aircraft. The A220's fuel system is designed to optimize fuel efficiency and distribution, with fuel management being crucial for maintaining balance and stability during flight. The weight of fuel on board varies depending on factors such as flight duration, distance, and payload.

The weight of cargo, passengers, and crew members also contributes to the overall weight and balance of the aircraft. The distribution of these loads affects the aircraft's center of gravity, which must remain within specified limits for safe flight. Airlines and operators conduct detailed calculations to ensure that weight and balance requirements are met for each flight.

Overall, the weight and balance component weights of the Airbus A220 encompass a range of elements, each carefully designed and managed to ensure the aircraft's safety, efficiency, and performance throughout all phases of flight.

This included calculating the center of gravity locations for each component of the aircraft. In addition, the sizing of the horizontal and vertical tail as well as landing gear configuration were obtained by using the initial weight and balance analysis performed. For the initial analysis, several assumptions were used to obtain the preliminary weight break down of the aircraft in order to locate the most aft and most forward C.G. location.

3.1.1 Gross Weight Estimation

For the optimized aircraft weight analysis is performed by taking the Empty weight and maximum take-off gross weight of the aircraft. For the initial analysis, several assumptions were used to obtain the preliminary weight of each and every passenger, cargo and fuel weight. Also, a bit of miscellaneous weights of the aircraft in order to calculate the optimized aircraft weight. The initial weight breakdown of the aircraft is tabulated below. Furthermore, the use of hydrogen will cause an increase in structural weight in the form of larger fuel tanks, insulators, and heat exchangers, this weight was not accounted for in the initial weight breakdown therefore a contingency was included in the take-off weight.

The estimated passenger the aircraft is intended to carry is 230 in a single-class cabin layout configuration. Assuming the average weight of the passenger as 91Kg that includes both clothing and baggage, we get the weight of the passengers to be (230*91) Kg = 21,850 Kg which is the $W_{payload}$ weight. Also, as per IATA and ICAO rules in order to manage an aircraft worth more than 200 passengers we need to have at 7 crew members and 2 Pilots, therefore taking the average crew weight as 91 Kg/person, we get the calculated crew weight as 819 Kg which is W_{crew}. Using the Empty weight estimation formula of:

$$W_e/W_0 = A\,W_0^C K_{vs}$$

Taking the values of the constant terms A and C as 1.02 and -0.06 from the "Empty Weight vs W_o" table and calculating all values we get W_e/W_o = 0.49.

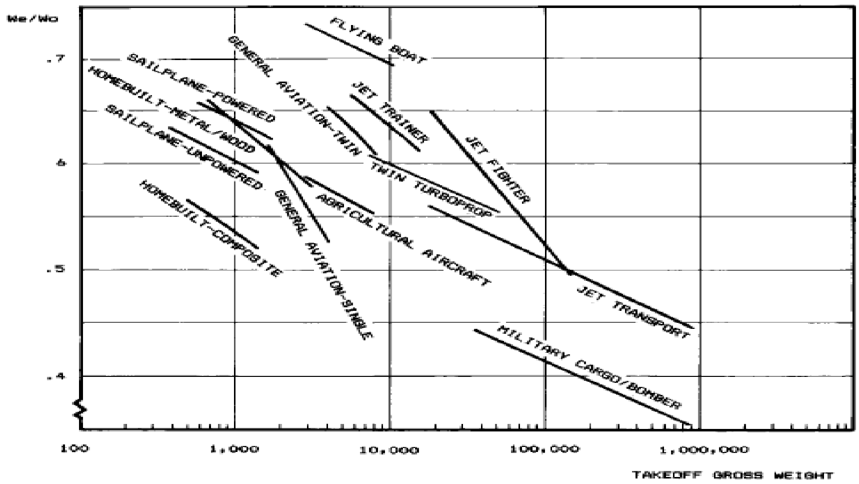


Fig. 3.1 Empty weight fraction trends.

Table 3.1 Empty weight fraction vs W_0

$W_e/W_0 = A\,W_0^C K_{vs}$	A	C
Sailplane—unpowered	0.86	−0.05
Sailplane—powered	0.91	−0.05
Homebuilt—metal/wood	1.19	−0.09
Homebuilt—composite	0.99	−0.09
General aviation—single engine	2.36	−0.18
General aviation—twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

K_{vs} = variable sweep constant = 1.04 if variable sweep
= 1.00 if fixed sweep

Figure 15: Empty Weight Fraction Trend

From the Empty weight fraction trend chart we are able to find the rough Take-Off Gross Weight for our optimized aircraft as shown below in the figure:

The Empty-weight fraction comes to be 0.49, which gives the Empty Weight fraction as 163225.23 pounds, after that utilizing the formula of W_f/W_o we get 0.3. Also, we get the Empty-Weight of the aircraft as 48,171.02 Kg

After application of the Gross-weight estimation formula as shown below:

$$W_0 = W_{\text{crew}} + W_{\text{payload}} + W_{\text{fuel}} + W_{\text{empty}}$$

We are able to get all the estimated take-off gross weight. There we need to find the Empty fuel wright and the Empty-Weight of the aircraft. By application of the theoretical formula to calculate the Fuel weight we have to follow the theoretical formula as shown below:

$$\frac{W_f}{W_0} = 1.06\left(1 - \frac{W_x}{W_0}\right)$$

We have found out that the ratio comes out to be 0.424, by using this imperial value we are able to calculate the Optimized Take-Off Gross weight which comes out to be 1,60,000 pounds or 72,574.77 Kg.

3.2 Fuel Fraction

Now from fuel fraction chart as per IATA and ICAO standards for transport jets:

1. Engine start and warmup – 0.990

2. Taxi – 0.990

3. Take off – 0.995

4. Climb – 0.980

5. Descent – 0.990

6. Landing, Taxi and Shut down – 0.992

Generally, at the end of mission, the fuel tanks are not completely empty, typically (6-8%) allowance is made for reserve and trapped.

Take-off gross weight (Wo) – 1,60,000 pounds

Conceptual LD estimation for jet aircraft:

Jet – 0.866 (mach) x $(LD)max$ = 0.866 x 19.25

Maximum speed of the aircraft = 0.86 Mach

Maximum range = 81000 nautical miles = 15001.2 km

Maximum estimated take-off gross weight = 72574.77 Kg = 159999.97 pounds

The basic form of the Breguet range equation is:

R=(vC). Isp. ln($W_{initial}W_{final}$)

Where:

- R = Range of the aircraft (distance travelled)
- v = Velocity of the aircraft (typically in meters per second)
- C = Fuel consumption rate (typically in kilograms per second)
- Isp = Specific impulse of the engine (a measure of engine efficiency, typically in seconds)
- $W_{initial}$ = Initial weight of the aircraft (including fuel)
- W_{final} = Final weight of the aircraft (including fuel)

Overall, fuel fraction of the aircraft.

Wf/WTo = 1.06 (1 - $W8WTo$)

Also, after we are able to calculate the optimized take-off gross weight, we are able to calculate the estimated fuel weight which comes out to be (Fuel-Weight Fraction * Optimized Take-Off Gross Weight) i.e. (0.424 * 72,574.77) Kg which results in 30,771.70 Kg. In order to calculate weight of fuel at different phases of flight we are using the imperial design method as shown below:

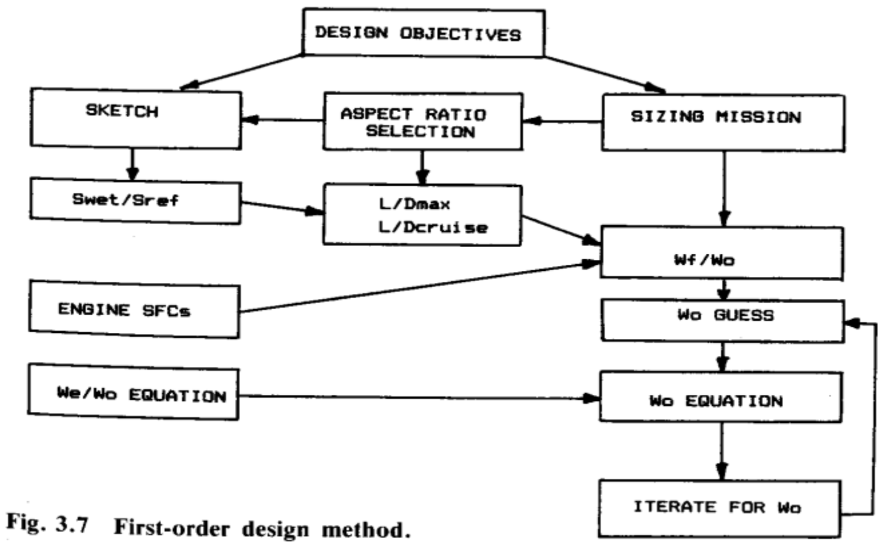


Figure 16: First-order Design Method

The calculated weight at different phases of the flight comes out to be as shown in the table:

Table 6: Summary of Aircraft Parameters: Engine Specifications

Mission Phase	**Fuel-Fraction**
Phase-1 (Engine Start)	0.990
Phase-2 (Taxi)	0.990
Phase-3 (Take-Off)	0.995
Phase-4 (Climb)	0.980
Phase-5 (Cruise)	0.650
Phase-6 (Loiter)	0.984
Phase-7 (Descent)	0.990
Phase-8 (Landing)	0.992

4. Design Configuration

4.1 **Fuselage**

The customized aircraft is a cutting-edge narrow-body aircraft designed for short to medium-haul routes. In this report, we explore the critical aspects of its fuselage, shedding light on its construction, materials, and unique features. This aircraft has been made by keeping in mind the toughness of the Fuselage Structure therefore this aircraft has Semi-Monocoque Design configuration**.** This aircraft features a semi-monocoqu**e** fuselage structure, a design that combines the best features of both monocoque and semi-monocoque designs.

Semi-Monocoque Explained:

- The term “semi-monocoque” refers to a structural configuration where the fuselage skin significantly contributes to the overall strength of the aircraft.
- In this design, the skin (outer shell) and internal frames (stringers and longerons) work together to distribute loads and stresses.
- The aircraft’s fuselage skin is primarily made of lightweight aluminium alloy panels, which are riveted to the internal frames.
- This semi-monocoque structure ensures a balance between strength and weight efficiency.

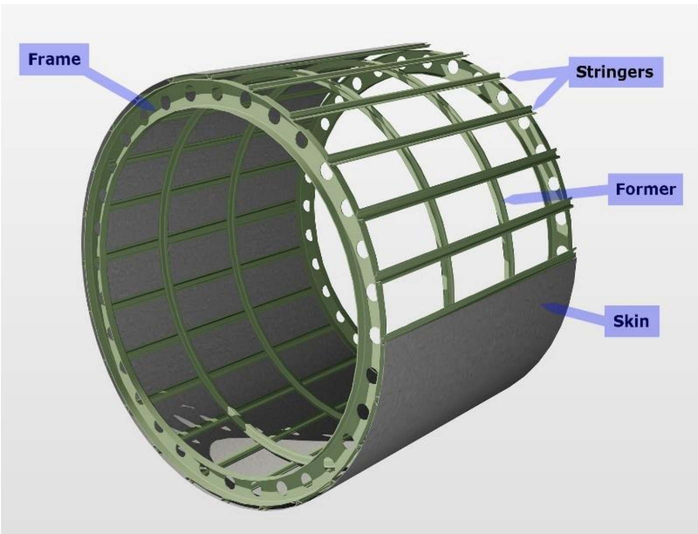


Figure 17: Semi-monocoque Structure portray.

The entire fuselage has been comprised of multi-type materials for enhancing durability, flexibility and prolonged structural fatigue life. General materials that will be used in the aircraft are listed below:

Aluminium Alloys:

- Aluminium remains the dominant material for the aircraft's fuselage.
- Aluminium alloys offer an excellent combination of strength, weight, and corrosion resistance.
- The aircraft will be utilizing advanced aluminium alloys to optimize performance.

Composite Materials:

- While aluminium is prevalent, composite materials also play a role.
- Specific components, such as fairings, radomes, and interior panels, incorporate composites.
- Composites, such as carbon fiber-reinforced polymers, provide high strength-to-weight ratios and allow for complex shapes.

This aircraft has been incorporated with Special Features of modern generation thus enabling maximum safety and comfort at all times irrespective of changing flight conditions and evolving flight parameters and wide change of modernisation ongoing in the modern aviation industry. The salient features that have been incorporated into the fuselage of this aircraft by keeping in mind the next generation upgradation and flexibility for wide range of customisation by the customers as per their needs might be for Business Use, Passenger operation, Freighter conversion and also for military Air Tanker.

The salient technical parameters of the fuselage of the aircraft have been mentioned:

Table 7: Parameter of Fuselage

Metric	Value
Overall Length	42.5 meters
Fuselage Maximum Diameter	3.92 meters
Cabin Length	34.3 meters
Height	13.0 meters

The features are enlisted below:

Wide Cabin for Passenger Comfort:

- This aircraft boasts a wider cabin than its competitors in the same category.
- Passengers enjoy increased shoulder room and spacious overhead bins.

Large Windows for Scenic Views:

- This aircraft will be featuring large windows enhance the flying experience.
- Positioned to maximize natural light, they reduce the feeling of confinement.

Advanced Aerodynamics:

- The fuselage shape is meticulously optimized for fuel efficiency and reduced drag.
- Smooth contours and a seamless wing-to-fuselage junction contribute to superior aerodynamic performance.

Quiet Cabin Design:

- Noise-reducing features and noise absorbent ceramic coatings make the aircraft's cabin quieter at all conditions.
- Improved insulation and minimized vibrations enhance passenger comfort.

The fuselage of this aircraft has been modified with optimal conditions for various door options. The Number of Doors included in the following numbers and locations are listed below:

This aircraft features a total of four passenger doors with options for 2 Door-Plugs. The normal door locations are enlisted below:

- Two forward doors (one on each side)
- Two aft doors (one on each side)
- Two Baggage Doors (Both in the Lower Starboard Side)

Additionally, there are **two emergency exits** (overwing exits) on each side of the fuselage. It will also be holding options for 2 additional door plugs for much more enhanced safety. The thoughtful design epitomizes efficiency, passenger satisfaction, and overall performance. As this aircraft continues to revolutionize regional travel, its fuselage remains a testament to engineering excellence.

A twin aisle cabin layout with a non-circular fuselage was chosen. A twin aisle configuration allows for a smaller cross-sectional diameter, which in turn reduces the overall drag experienced by the aircraft. Furthermore, a non-circular configuration was chosen to remove unnecessary space underneath the fuselage as the mission specification doesn't require the carrying of cargo along with passengers and baggage. This in turn reduces overall structural weight while also minimizing drag. The cabin has been designed for optimal use of space and the seating has been calculated to accommodate 230 passengers at maximum with full Single-class Economy seat configuration. This configuration can be modified to both Business Class and Premium economy depending on the needs and requirements of the customers. The detailed planned layout of the cabin is given below:

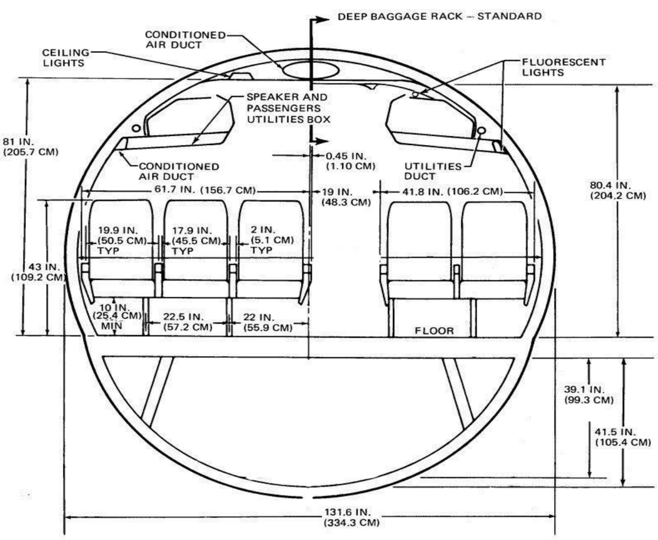


Figure 18: Entire vertical-cross section of the cabin layout.

<u>Modified Cabin Length and Width:</u>

The cabin length of the aircraft is 332.6 centimeters (or approximately 3.33 meters).

The cabin width measures 334.3 centimeters (or approximately 3.34 meters).

Fuselage Length:

The cabin layout can be visualised in multiple viewports with the entire and full-service carrier configuration as shown below:

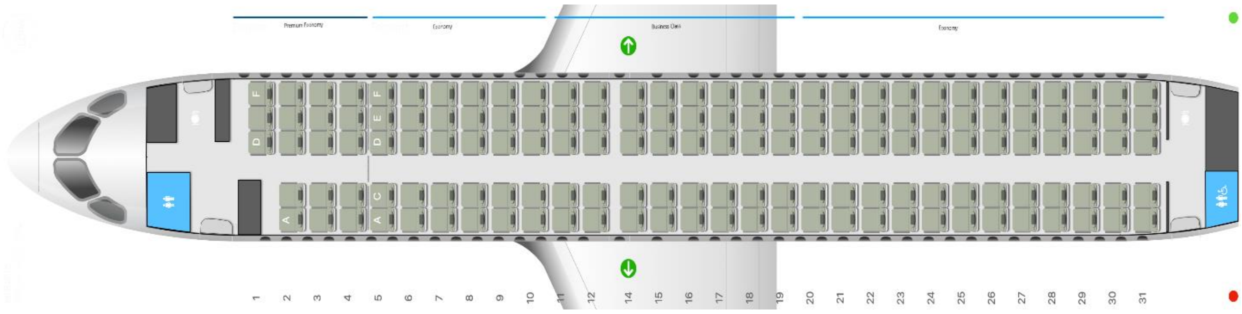


Figure 19: Seating Arrangement for Full-Service carrier.

Here, the entire seating arrangement as per traditional norms has been completely modified and overhauled to accommodate and utilise maximum space as much as possible. From the front part of the fuselage, we are getting the configuration as:

Premium Economy Class ----→ Economy Class --→ Business Class --→ Economy Class

A detailed outlook to the cabin width with respect to a 6 feet human is visualized below which gives us a clear idea about the spacious cabin that can easily withstand the size of a full-grown human inside it making it an all-range carrier and suitable for all airlines to use in their short to medium-haul service.

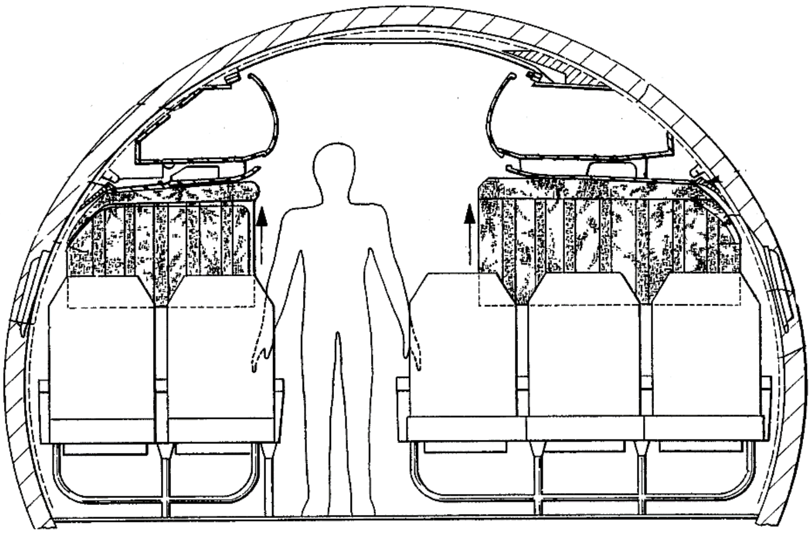


Figure 20: Cabin Layout Interpretation

There are 2 lavatories installed at both ends of the cabin to accommodate passenger ease and comfort, with enough space for a person to easily complete their nature's call. Also. the lower deck cross-sectional shape also allows for two LD3-47 unit loading devices to be fitted next to each other. As a result, it is expected that there will be more space available in the lower deck for the fuel storage and heat exchanger devices. In addition, compared to an oval shape, this custom cross-sectional shape will increase the passenger comfort due to the extra overhead space available.

4.2 **Airfoil Selection and Characteristics**

Detailed analysis and study of airfoil gave us the NACA 66(4)-221 as the most suited airfoil for our operational needs as this is the airfoil which plays a crucial role in shaping the performance of the Airbus A220. Let's explore why this specific airfoil is well-suited for this aircraft type:

4.2.1 Supercritical Design:

- The NACA 66(4)-221 airfoil belongs to the supercritical airfoil family.
- Supercritical airfoils are optimized for transonic and high-subsonic speeds.
- Modern airliners, including the A220, operate in this speed regime.

4.2.2 Key Features:

- **Max Thickness**: The airfoil has a maximum thickness of 12% at 45% chord.
- **Max Camber**: It exhibits a maximum camber of 1.3% at 47.5% chord.
- These features strike a balance between lift generation, drag reduction, and structural efficiency.

4.2.3 Advantages for our Aircraft:

- **Fuel Efficiency**: The NACA 66(4)-221 airfoil minimizes drag, contributing to the A220's impressive fuel efficiency.
- **Stability**: Its design ensures stable flight characteristics, especially during takeoff and landing.
- **Transonic Performance**: The supercritical shape allows the A220 to handle transonic airflow speeds over the upper wing surface.

4.2.4 Potential Modifications:

- While the NACA 66(4)-221 airfoil is well-matched for future upgradation as there's room for improvement.

In summary, the NACA 66(4)-221 airfoil strikes a balance between efficiency, stability, and transonic performance, making it an excellent choice for our aircraft. As technology evolves, further refinements may enhance its capabilities.

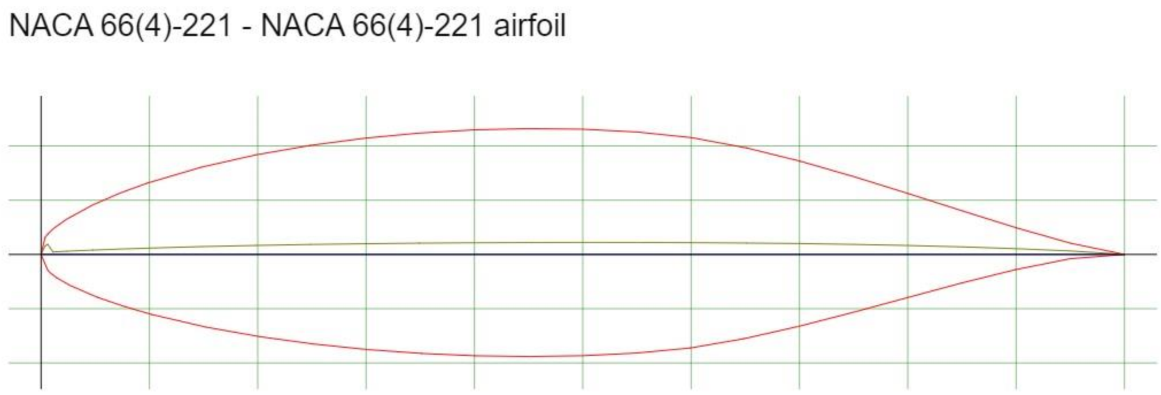


Figure 21: NACA 66(4)-221 Airfoil layout as per Airfoil Tools

Advantages and specific reasons for why we have chosen the NACA 66(4)-221 Airfoil for our customized Aircraft design.

The NACA 66(4)-221 airfoil, like other NACA airfoils, offers several advantages when used in aircraft design. Let’s explore its benefits:

1. Efficient Lift and Drag Characteristics: The NACA 66(4)-221 airfoil is carefully designed to provide a good balance between lift and drag. Its shape allows for efficient generation of lift while minimizing drag, which is crucial for overall aircraft performance.
2. Reduced Drag: The airfoil’s streamlined shape helps reduce parasitic drag during flight. Lower drag means the aircraft requires less power to maintain a given speed, resulting in better fuel efficiency.
3. Stability and Control: The NACA 66(4)-221 airfoil exhibits stable aerodynamic behavior, making it suitable for various flight conditions. Pilots can rely on predictable handling characteristics during takeoff, landing, and maneuvers.
4. Versatility: NACA airfoils, including the NACA 66(4)-221, have been successfully used in wings, tail sections, propellers, and helicopter rotors. Their versatility makes them valuable for both general aviation and military aircraft.
5. Material Choices: The NACA 66(4)-221 airfoil can be manufactured using different materials, such as aluminum, carbon fiber, or epoxy glass. Material selection affects structural strength, weight, and natural frequency, allowing designers to tailor the airfoil to specific requirements.

A graphical representation of the parameters of the CL and CD change with respect to Angle-Of-Attack will give us an estimated idea about the airfoil characteristics in much briefer.

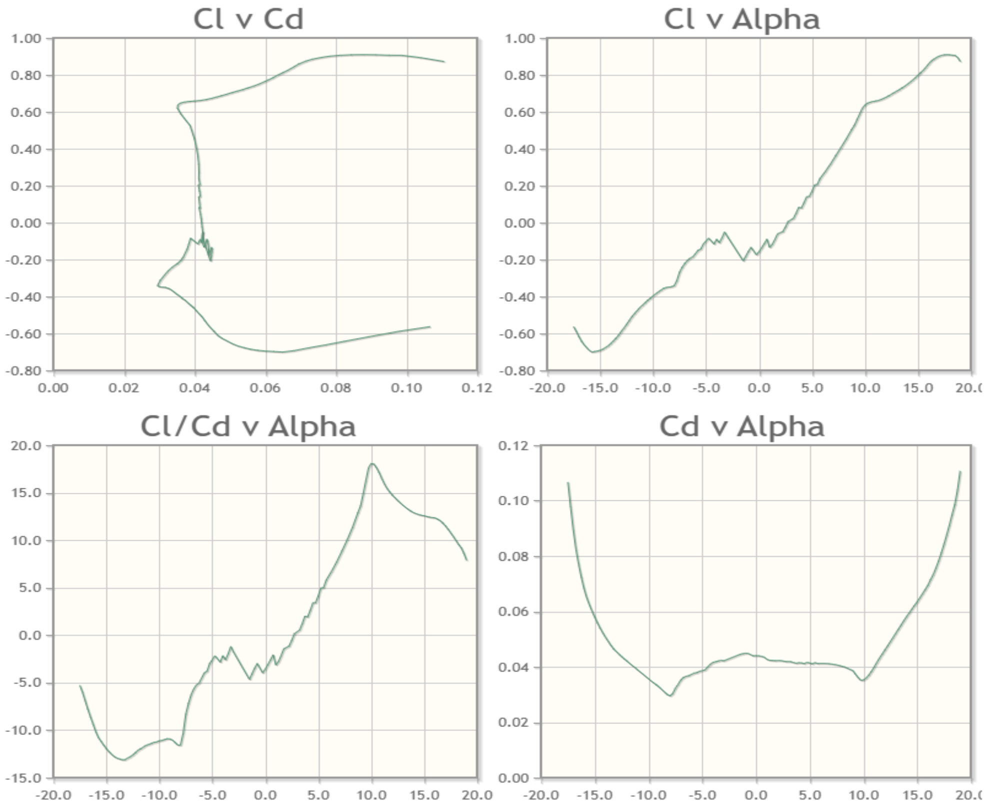


Figure 22: NACA 66(4)-221 airfoil characteristics graph as per Airfoil Tools

4.3 Wing Location and Configuration

Mid-wing and Low-wing configurations were considered in the design of this aircraft. Below is a short description of each setup:

<u>Mid-Wing Configuration:</u> The wings are exactly at the midline of the airplane. Mid-Wing airplanes are very well balanced, and their design allows for a large control surface area (portions of the plane involved in steering). This configuration is very manoeuvrable but not is not as stable as high wing airplanes.

<u>Low Wing Configuration:</u> The wings are below the midline of the airplane. This configuration is more stable than mid-wing airplanes but not as much as high wing airplanes. Although this is not a desired characteristic, low-wing airplanes are more manoeuvrable than high-wing airplanes. In addition, the lower height of the wings allows for easy engine maintenance if the engines are mounted under the wing.

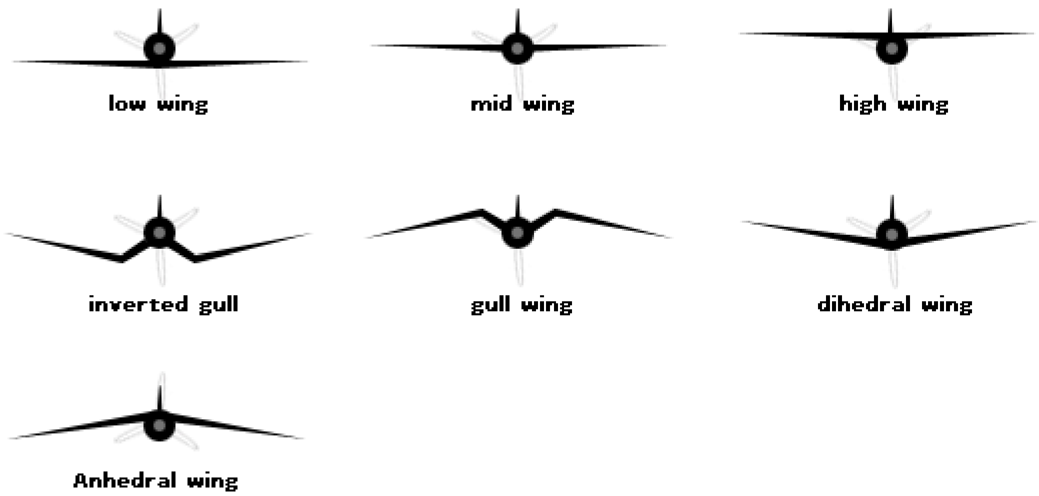


Figure 23: Wing Location Configuration

Wing Placement: A mid-wing dihedral configuration was chosen as it is the industry standard with most transport aircraft having this configuration. Mid-wing provides the structural components like the spars of the wing to go through the central part of the fuselage. This makes integration with cargo and passengers optimal as the top half of the fuselage is preserved for passengers only. In addition, the engines can be serviced easily if they are mounted under the wing as they offer enough ground clearance for larger engines. Currently, the engines are placed near the empennage, but low-wing configuration would be beneficial if the engines are chosen to be moved under the wings. In that case, to account for clearance, more wing dihedral may be added. For this aircraft by minute specific calculation and detailed analysis of studies conducted by Airbus we found that a Mid-Wing Dihedral configuration will much suit the type in order to accommodate a large size engine for proper ground clearance Aerodynamic Swept Low Wing which the modified aircraft features a swept low wing design. Here is why it matters:

- **Efficiency:** The swept wing reduces drag during flight, enhancing fuel efficiency.
- **Stability:** The low-wing configuration provides stability and better lift distribution.
- **Winglets:** The wingtips incorporate winglets, which further reduce drag and improve overall performance.
- **Carbon Composite Wing:** The aircraft wing is constructed using advanced carbon composite materials making it:

- **Lightweight:** Carbon composites are lighter than traditional materials, contributing to fuel savings.
- **Strength:** They offer high strength-to-weight ratios, ensuring structural integrity.
- **Optimized Aerodynamics:**
- The wing design is meticulously optimized for enhanced:
- **Aspect Ratio:** The wing's aspect ratio balances lift efficiency and maneuverability.
- **Wing Loading:** Proper wing loading ensures optimal performance during takeoff and landing.
- **High-Lift Devices:** Flaps and slats enhance lift during critical phases of flight.

4.3 High-Lift Systems

In order to bridge the gap between the maximum coefficient of lift in the clean configuration and the maximum coefficient of lift needed for take-off and landing, high lift devices will be added to the aircraft. These devices, while adding the needed lift for certain flight conditions, also add weight, complexity, and cost to the aircraft and must be designed with this in mind. The two pertinent areas of the aircraft that benefit the most from high lift devices are the leading and trailing edges of the wing. This is because the simplest methods to increase the lift of the aircraft is by increasing wing area and wing camber. Trailing-edge flaps are critical to almost every commercial aircraft in use today. They have different deployment settings during take-off and landing and many planes would not be able to do either without them. Fowler flaps are the most widely used flap design in the commercial industry due to their relative simplicity and their increase in wing area and camber. Fowler flaps extend out on tracks and often have a series of slots to add energy to the airflow to keep the flow attached over the flaps. The most common of which are the single-slotted, double-slotted, and triple-slotted Fowler flaps. The more slots there are increases the total lift and drag produced, but also greatly increases weight, complexity, and cost. The most desirable quality of these flaps is that different flap settings are ideal for different airplane configurations. For example, at low flap deployment levels, the Fowler flaps extend from the wing and increase wing area and thereby lift considerably, while only creating a slight increase in drag which is ideal for a take-off configuration. At high deployment levels, the flaps extend more in a downwards motion and create a little higher lift, but also a large increase in drag which is ideal for a landing configuration. The trailing-edge high lift system was sized in an iterative process by choosing an initial flap to wing area ratio and comparing the resulting change in coefficient of lift to the required change in coefficient in lift for both landing and take-off conditions.

Looking at the leading edge of the wing, the two most viable designs to increase lift are Krueger flaps and slats. Krueger flaps are typically cut-out of the cruise-wing geometry and extend forwards during take-off and landing to increase wing camber and area. The folding, bull-nosed Krueger flaps can provide a larger maximum coefficient of lift than slats can, however, they also have a smaller lift-to-drag ratio during take-off than slats do. When deployed, Krueger flaps create a very blunt leading edge that becomes ideal to be placed in the thicker (inboard) portions of the wing. Slats are simple leading-edge devices that, when deployed, increase the camber of the wing while keeping the leading edge of the airfoil thin. Doing so assists in lowering the stall speed and decreasing the take-off and landing ground roll of the aircraft. The differences in deployment methods are shown in the figure below.

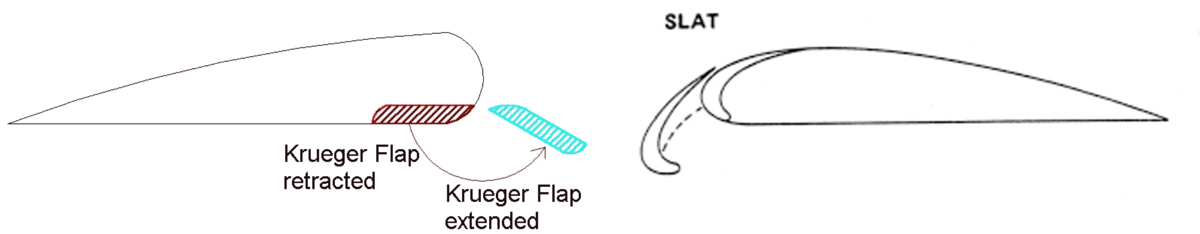


Figure 24: Diagram of Krueger Flaps and Slats

4.4 Engine Placement

The engines were placed under the wings closer to the Wing-Fuselage fairing, one in the Starboard side and the other in the Port side. The mid-wing dihedral configuration allows the large diameter turbofans to be installed easily with proper ground clearance for optimal movement of the aircraft and also facilitates maintenance, it also helps for the placement of future larger bypass ratio engines.

The Engine has been placed by keeping in mind enough safe space is there for Jet-Blast to evade and to not damage the airport nearby surroundings with enough room available for the pilot to move and keep a safe distance from the person on ground. This is pictorially shown in the image below:

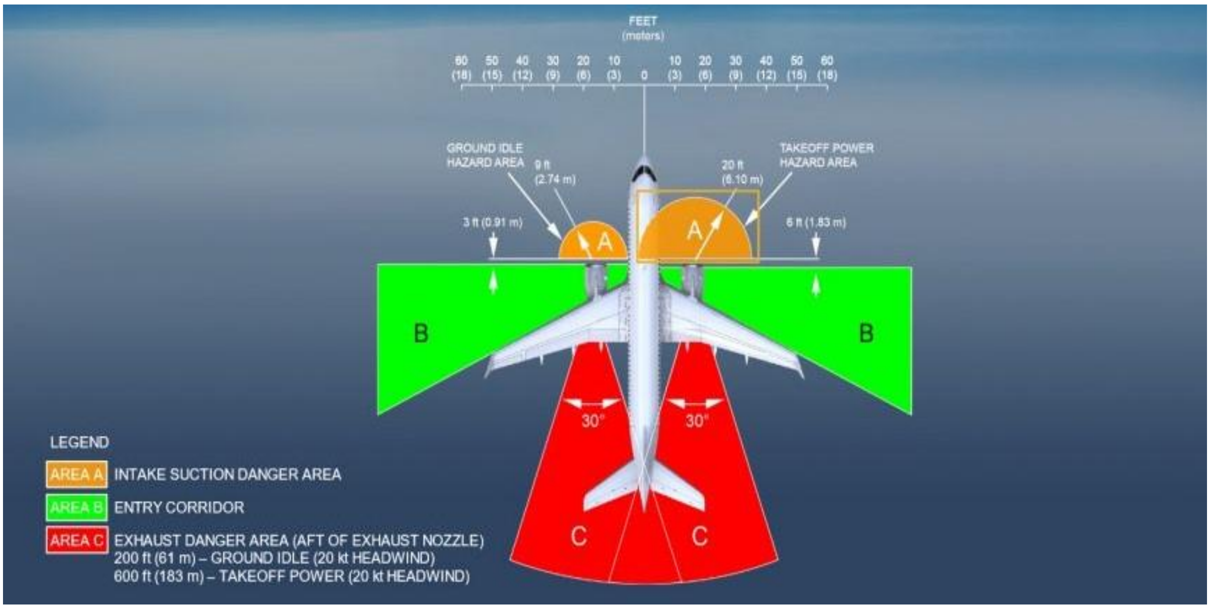


Figure 25: Engine Ground Clearance hotspot areas.

4.5 Tail Configuration

The tail configuration chosen for the aircraft is a conventional tail design this is due to the following reasons:

Conventional Tail: This configuration consists of a vertical stabilizer (commonly known as the tail fin) and a horizontal stabilizer. When viewed from behind, it forms an "L" shape. The horizontal stabilizer provides pitch control, while the vertical stabilizer provides yaw control.

This design also provides adequate stability and control with the lowest structural weight.

A Conventional tail design is chosen specifically for this aircraft due to the following reasons The conventional tail design offers several advantages for civil jet aircraft like:

4.5.1 Stability and Control:

- The conventional tail configuration consists of a vertical stabilizer (fin) and a horizontal stabilizer (elevator).
- The fin provides directional stability, preventing yawing motions during flight.
- The elevator controls pitch, allowing the aircraft to maintain a desired angle of attack.

4.5.2 Predictable Handling:

- Pilots are familiar with conventional tail designs, making them easier to handle during take-off, landing, and manoeuvres.
- The A220-300's conventional tail ensures predictable responses to control inputs.

4.5.3 Reduced Risk of Stalling:

- The horizontal stabilizer's position above the fuselage contributes to natural pitch stability.
- This stability minimizes the risk of stalling during critical flight phases.

4.5.4 Aerodynamic Efficiency:

- The conventional tail design optimizes airflow around the tail surfaces.
- The fin and elevator work together efficiently, enhancing overall aerodynamics.

4.5.5 Space for Systems and Components:

- The conventional tail frees up space within the fuselage for avionics, hydraulics, and other systems.
- It allows for clean integration of components, simplifying maintenance.

4.5.6 Community-Friendly Noise Reduction:

- Quieter operations benefit both passengers and communities near airports.
- The conventional tail contributes to its half the noise footprint compared to previous generation aircraft.

4.6 Landing Gear Configuration

The landing gear used for the aircraft has a robust design and a wide popular outlook in the aviation market as the most used landing gear type for the Airbus A330 variant and we have chosen to move forward with that same landing gear type and configuration in order to accurately manage our ground height clearance and handle the immense load at the time of landing and standing at the gates. The landing gear is a critical component of any aircraft, impacting safety, performance, and maintenance. For the aircraft, the choice of landing gear is carefully considered due to the reasons as:

Figure 26: Safran System Landing Gear Main Hub (Port Side)

The main and nose landing gears for the A330neo are designed by Safran Landing Systems.

This landing gears has features like:

4.6.1 Robust Design and Reduced Maintenance:

- Safran’s landing gear design ensures robustness and reliability.
- Reduced maintenance actions lead to cost savings for airlines.

4.6.2 Optimized Weight:

- The landing gear is optimized for weight, contributing to overall fuel efficiency.
- Weight reduction enhances the A330neo’s performance during take-off and landing.

4.6.3 Use of New Materials:

- Safran incorporates advanced materials meeting environmental requirements.
- These materials enhance durability and reduce environmental impact.

4.6.4 Operational Flexibility:

- The landing gear is tailored for flexible operations.
- It allows the aircraft to serve diverse routes and adapt to changing market demands.

4.6.5 Noise Reduction and Community-Friendly Operations:

- The landing gear contributes to its community-friendly noise reduction features.
- Quieter landings benefit both passengers and airport neighbours.

The location and height and other general dimensional characteristics of the landing gear have been given in the table below:

Table 8: General Dimensional characteristics of the Landing Gear

ACCESS	DISTANCE			
	AFT OF NOSE	FROM AIRCRAFT CENTERLINE		MEAN HEIGHT FROM GROUND
		LH SIDE	RH SIDE	
On NLG leg	8.17 m (26.80 ft)	On centerline		1.40 m (4.59 ft)
On left MLG leg	32.1 m (105.31 ft)	5.34 m (17.52 ft)		1.50 m (4.92 ft)
On right MLG leg	32.1 m (105.31 ft)		5.34 m (17.52 ft)	1.50 m (4.92 ft)

5. Propulsion Systems

5.1 Propulsion Requirements & Engine Selection

The thrust to weight parameter was computed using the sizing diagram populated in Figure 7 and was found to be 0.35. Using Roskam part one, the takeoff weight was determined to be 581,124 lbs, resulting in a thrust requirement of 203393.4 lbf. A twin-engine arrangement was desired for the configurations of this aircraft, requiring each engine to produce roughly 101,500 lbf pounds of thrust. A trade study of a range of high-performance engines currently used in the market were analyzed, the table below summarizes findings:

Table 9: Engine Selection Trade Study

Engine	Max Thrust [lbf]	Weight [lbf]	Diameter [in]	Pressure Ratio	Bypass Ratio
GE90	82 - 97,000	19,000	123	40:1	8-9:1
PW400	90-99,000	16,260	112	34-42:1	5.8-6.4:1
RR Trent 800	77- 93,00	14,000	110	34-40:1	6.4:1
GE 2B67B	66,500	12,397	104.7	54:1	8:1

Per the requirements of the design, one of the primary considerations is to reduce overall operational costs for a reference distance of 700 nm. With an aim is to reduce fuel emissions by 75%, NOx by 90% and noise by 60% relative to aircraft in 2000. Alternative fuels are considered in a later section to further address this issue. From the analyzed engines it can be seen that the GE 2B67B proves to be the best option. The high-pressure ratios raise the inlet temperature of the fuel, which consequently increases net thrust and decreases specific fuel consumption. The GE 2B67B also has the highest bypass ratio on the table shown above, high bypass ratios ensure that large amounts of air are passed through narrow nozzles which increase thrust whilst using the same amount of fuel. According to GE, the combination of these two parameters provides a 10% improvement in fuel burn when compared to the GE90-115B, a significant decrease when considering roughly 19% of an airlines operational cost is fuel.

Table 10: Summary of Aircraft Engine Specifications

Engine Specifications	
GE 2B67B	
Max Thrust [lbf]	66,500
Weight [lbf]	12397
Length [in]	170
Diameter [in]	104.7
Pressure Ratio	54:1
Bypass Ratio	8:1

The downside of using the GE 2B67B comes from the large fan diameters as well as the considerably heavy weight. Thus, to mitigate ground clearance issues and reduce the number of debris entering the engines, it was decided that mounting the engines on the rear of the fuselage would be the best route. Another option would be to mount the engines atop the wings, structural implications of these two configurations are currently being explored. The weight of these engines was considered when computing the initial sizing and are included in the takeoff weight.

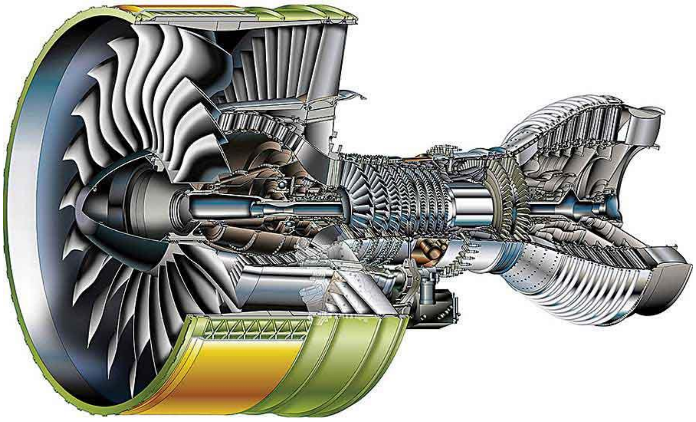


Figure 27: GE 2B67B.

5.2 Alternative Fuel Source Candidate - Hydrogen

The industry standard for aircraft fuels are different blends of kerosene with thousands of other additives like anticorrosives, biocides, and icing inhibitors. However, the use of these fuels causes a negative impact on the environment. The global aviation industry produces 2% of all human induced carbon dioxide, while aviation produces 12% from all transport sources.

Furthermore, with depleting oil reserves, stringent ICARE goals, and the overall greenhouse effect, it is imperative that alternative fuel sources be explored. One of these is a renewable energy source, compressed hydrogen. Hydrogen is a high-energy clean burning fuel whose main combustion byproduct is water vapor and trace amounts of nitrogen oxides. Furthermore, converting already existing turbofan engines is relatively simple to do without drastically changing the engine configuration.

Hydrogen gas has a significantly lower density than Kerosene, thus storing it in this state is not feasible due to weight and size restrictions. When compressed and cooled, the gas converts to a liquid which significantly increases the density (0.089 g/L for gas, 71 g/L for liquid) and reduces required tank size. However, this liquid state must be converted into a gas for efficient combustion. This can be accomplished by installing heat exchangers at various location of the engine. These heat exchangers aim to raise the overall temperature of the flowing fuel, converting the fuel to the desired state while also reducing the specific fuel consumption (SFC). This lowered SFC in conjunction with the higher energy density of Hydrogen results in a 64.7% reduction in SFC when compared to Kerosene. Hydrogen engines also run cooler than conventional kerosene engines which results in a longer life for the engines as components won't face stringent heat wear. The final benefit of using hydrogen comes from the saved cost in the form of carbon tax, the renewable nature and relatively harmless emissions make it exempt from such levies. Further analysis is being done on the safety implications using this fuel as well as a trade-off analysis between the volume/weight of hydrogen compared to the volume/weight of kerosene for the reference mission of 700 nm.

5.3 Detailed Design and Analysis: Hydrogen Storage

As discussed in the propulsion section, two different fuel options are considered to be implemented for our design. This approach is taken to minimize costs for the goal of slowly transitioning into long-range commercial transport using hydrogen fuel for the sake of a sustainable future for aviation. Moreover, this detailed design study focuses solely on the storage of hydrogen. However, the effects of the modifications to implement hydrogen on the original design are important to consider and are briefly discussed below.

It is realized later in above Section that hydrogen is stored in the liquid hydrogen state and at cryogenic temperatures. Due to the physical properties of cryogenic liquid hydrogen and its highly explosive nature, it is ideal for it to be stored in the fuselage, rather than the wings. This requirement causes one major design change — location of fuel tanks. In order to minimize further major changes, the tanks are simply placed in the front and the rear of the fuselage with the goal of preserving the existing seating layout within the fuselage.

Furthermore, there are few other changes caused by the above modification which are noteworthy. The wings are not held down by fuel or engines during flight, so they require extra structural reinforcements. This increases empty operating weight in comparison to the original kerosene design. However, hydrogen fuel weighs much less than kerosene so the maximum takeoff weight actually decreases. These effects are only to name a few and several other changes are expected, which will be considered in the future.

5.3.1 Sizing and Structures

As there are two fuel tanks to be implemented within the airplane, each fuel tank is sized to hold $4196.9ft^3$ of hydrogen. According to Gomez, elliptic domes with a major to minor axis ratio $\frac{a}{b} = 1.66$are the best compromise of weight and tank length. This conclusion is used in the sizing along with the diameter restriction from the fuselage cross section. The following numbers for tank length and diameter are obtained using the above relations:

Table 11: Tank and Fuselage Dimensions

Tank Length	39.19 ft
Tank Diameter	10.73 ft
Fuselage Diameter	12.86 ft
Fuselage Length	139.43 ft

The sizing numbers conclude that there is a few inches of vertical clearance and few feet of horizontal clearance within the tank and fuselage. This should be sufficient for the structures within the fuselage where the tanks will be placed and will require mounting. Furthermore, the tanks take up about 80 ft along the length of the fuselage, which is one-third of the fuselage length.

Shell

The figure below shoes design of the tank shell developed in Solidworks. It can be seen that the tank is cylindrical rather than elliptic, which is done to minimize manufacturing costs. The optimized a/b ratio of 1.66 is still applied. Furthermore, the shell is 2mm thick and reinforced by a frame as discussed below. The material of the shell is discussed in a later section as well.

Frame

The hydrogen tanks are designed to withstand the pressure required for cryogenic storage. As cryogenic storage is chosen over high-pressure storage, tanks can be sized as low-pressure vessels. In addition to the skin, the tanks include a two-layer frame structure that consists of outer rings and inner longitudinal stringers. The tank skin is attached to this frame.

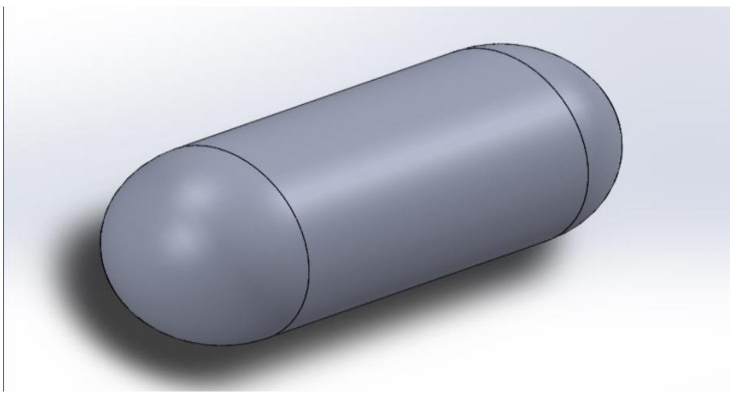


Figure 28: Hydrogen Tank Shell - Iso View

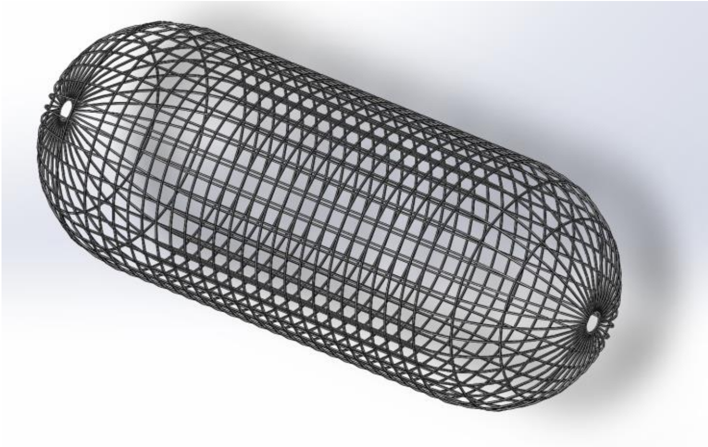


Figure 29: Hydrogen Tank Frame - Iso View

Table 12: Hydrogen Tank Frame Dimensions

Parameter	Value
Frame Length	39.19 ft
# of Outer Rings	24
# of Longitudinal Stringers	36
Stringer Thickness	1.0 in
Stringer Width	1.0 in

The frame matches the shape of the tank skin and has two domes and a cylindrical section. There are 24 outer rings and 36 longitudinal stringers. This is initial frame design is subjected to change as per FEA results, the structure can be altered to make it stronger in the case of failure.

The tank frame can be seen in top and side view as well:

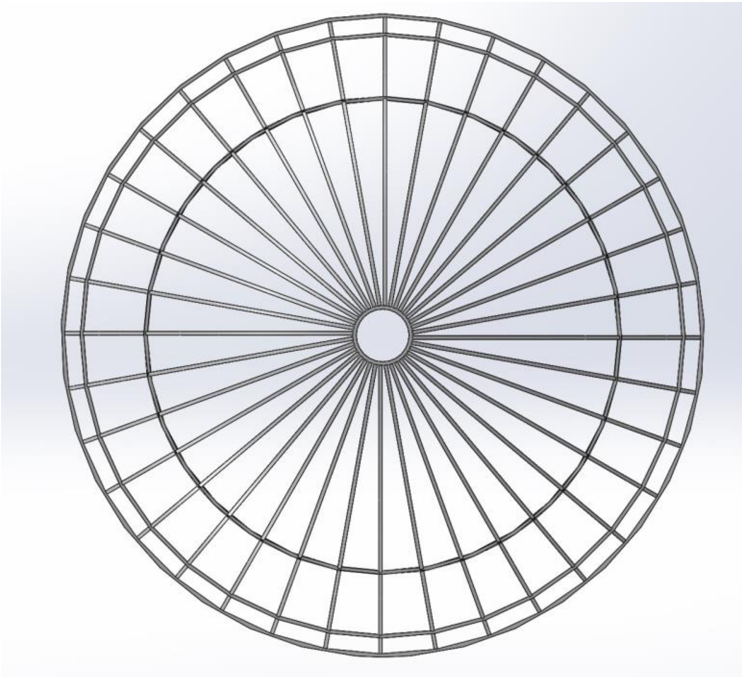


Figure 30: Hydrogen Tank Frame - Side View

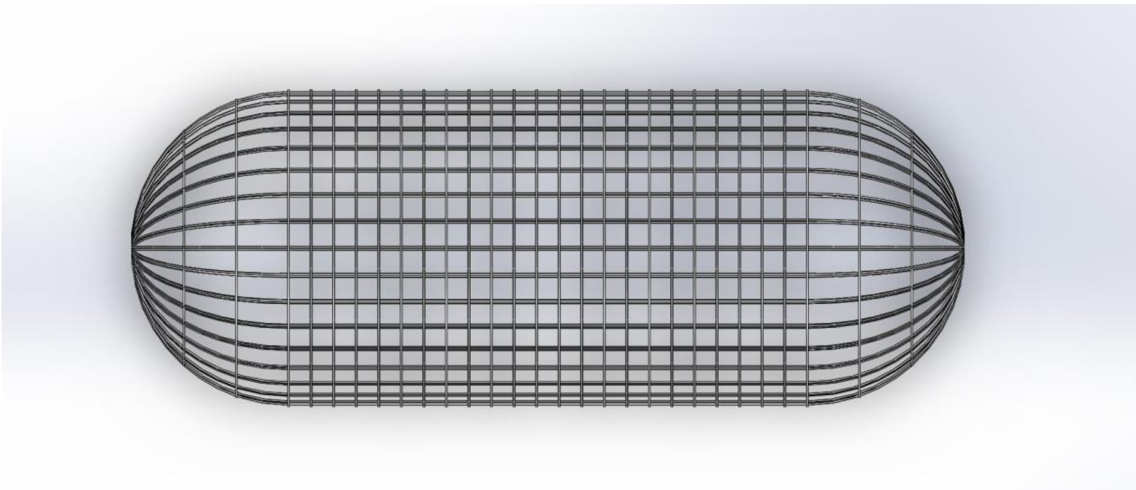


Figure 31: Hydrogen Tank Frame - Top View

Storage

The main problem with storing hydrogen on aircraft is its extremely low density compared to normal jet fuel. Storing a sufficient amount of hydrogen calls for large storage tanks that increase the weight and decrease the usability of an aircraft. So, in order to make hydrogen propulsion practical, the storage method must be able to increase the density of hydrogen to decrease the size and weight of the storage tanks.

The two main methods of hydrogen storage are high pressure storage and cryogenic storage. The high-pressure method stores hydrogen as a gas at pressures up to 10,000 psia and is the most conventional and widely used storage option.

As storage pressure increases, the density of hydrogen also increases, and this pattern is shown in the figure below.

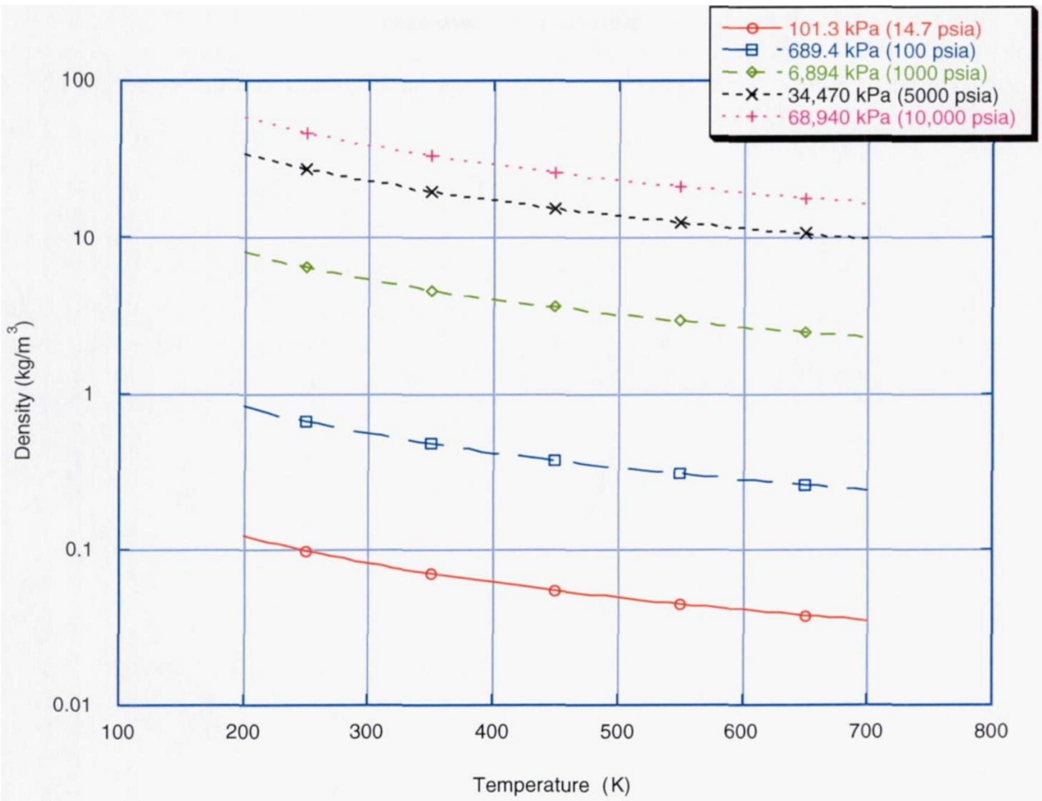


Figure 32: Hydrogen Density at Various Pressures and Temperatures

It is important to note that as pressure increases, the storage tank thickness must be increased to withstand the applied stresses. It also needs expensive and heavy systems equipment to maintain the tanks at high enough pressures to hold enough hydrogen for the 3500 nautical mile mission requirement. This, along with the fact that high pressure storage is quite dangerous and could lead to loss of life, are the reasons cryogenic storage has been chosen for our design.

The cryogenic storage system allows for a reduction in tank weight and volume by storing the hydrogen as a liquid. Liquid hydrogen has a density about 70 times larger than gaseous hydrogen at standard atmospheric pressure, but it must be kept below the boiling temperature of Hydrogen of 21 Kelvin. To further increase the density above that of liquid hydrogen, the hydrogen can be stored as a half-and-half mixture of solid and liquid hydrogen, called slush hydrogen, which has a density 80 times larger than gaseous hydrogen. The effect that the temperature and pressure of the hydrogen is stored at is shown below, indicating there is no need for extreme pressurization of the tanks.

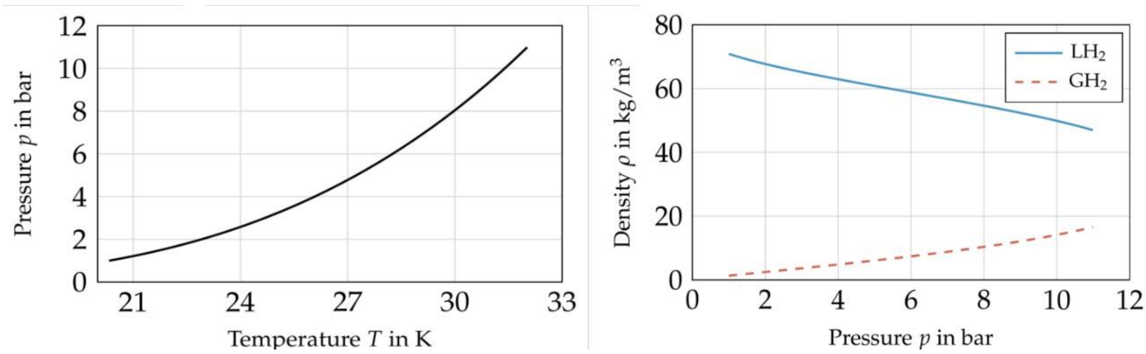


Figure 33: Cryogenic Temperature, Pressure, and Density Values

Cryogenic storage maximizes the amount of hydrogen to be stored in the smallest possible volume and weight, however, it does not come without its difficulties. Because it must be maintained at cryogenic temperatures, an airtight insulation system must be used to reduce the boil-off rate. Cryogenic hydrogen also has a limited lifetime due to the boil-off rate and will most likely have to be produced on-site. The fuel tanks must also be constantly sealed and pressurized (around 25 psia) to reduce boil-off rates and ensure no air or liquid gets inside which would instantly freeze and clog fuel lines. That being said, the only two significant systems involved in storing the cryogenic hydrogen are the tank and insulation.

Recent research in cryogenic storage has also shown potential for laminar flow control by use of the sensible heat of the stored fuel to cool aerodynamic surfaces. The cryogenic temperatures allow for a cooling of the boundary layer on the wings and nacelles to delay the transition to turbulent flow and reduce the frictional drag. Although the implementation of surface cooling increases the aircraft operating empty weight through addition of heat exchangers, pumps, and control subsystems, the laminar flow control has shown to be a feasible and cost-saving application of cryogenic hydrogen storage.

5.3.2 Materials and Insulation

Materials

In order to safely carry liquid and gas hydrogen, the tank needs to be made of materials that have high strength, high fracture toughness, high stiffness, low density, and low permeation to liquid and gas hydrogen. A trade study was performed between aluminum, steel, titanium, and composites to determine the best material for the tank. The material and mechanical properties are listed in Table.

The material selection process considers the properties listed in Table as well as the cost of the material and its manufacturability. 304L steel was identified as a potential material; however, its material properties were undesirable due to its high density and low yield strength compared to other materials. Also, steel performance is unreliable in cold temperatures, which is a concern because the hydrogen needs to be stored in cool tanks. TC4 titanium was also analyzed, and its material properties were in a desirable range with a low density, high yield strength, and low thermal conductivity. However, the raw material and manufacturing cost of titanium is unfavourable. Composite materials showed favorable properties such as in the density, yield strength, and thermal conductivity. Despite the potential for a significant reduction in weight, the costs of production and development are high for composite materials compared to conventional metals. Also, there are potential safety concerns about permeation with hydrogen and being able to identify fractures.

After identifying all the potential materials, the 209X series aluminum was selected as the choice material. The 209X series advanced aluminum is an ideal material because of its low density and its yield strength is in a desirable range. Aluminum is also reliable in cold temperatures. Although aluminum has a high thermal conductivity, it will be insulated with a composite material to keep the tanks cool. Aluminum is also affordable to acquire and manufacture.

Table 13: Properties of Select Materials

Material	Density	Yield Strength	Thermal Conductivity
Aluminum (209X)	2.63 g/cm^3	470 MPa	164 W (m K) 1
Steel (304L)	7.93 g/cm^3	410 MPa	15 W (m K) 1
Titanium (TC4)	4.5 g/cm^3	825 MPa	7.96 W (m K) 1
Composite	1.53 g/cm^3	1900 MPa	0.00675 W (m K) 1

Insulation

LH2 needs to be stored at low temperatures to keep the chemical composition as a saturated liquid/gas mixture.

However, during various stages of flight, the tank is exposed to numerous heat sources from its surroundings, causing variations in tank pressure and temperature. This in turn causes LH2 boil off rates to increase. Acceptable boil off rates for aircraft are in the order of 0.1% by weight per hour or less. Furthermore, to minimize the operating empty weight of the aircraft, low density foams such as aerogels and multi-layer insulation (MLI) are the most desirable. MLI consists of alternating layers of low conductivity spacer and low emissivity foil and offer thermal performance an order of magnitude better than foam. Unfortunately, this performance degrades rapidly for pressures higher than 100 mPa, posing an unnecessary safety risk. The proposed tank uses polyurethane foam for insulation material.

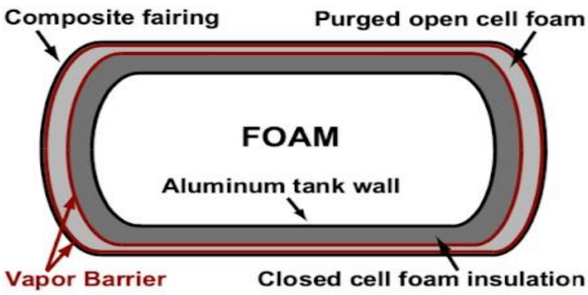


Figure 34: Fuselage and Tank Layers

The insulation can be applied to the interior or exterior of the tank wall. If the insulation is placed outside the tank, diffusion of gaseous hydrogen will prevail and prevent effectiveness. Furthermore, the insulation is also more susceptible to mechanical damage as the tank volume increases and decreases at various stages of flight. Lastly, since the liquid hydrogen is at cryogenic temperatures, direct exposure to the tanks can cause some undesirable effects to the structural integrity of the tank. Thus, interior insulation is chosen as shown above.

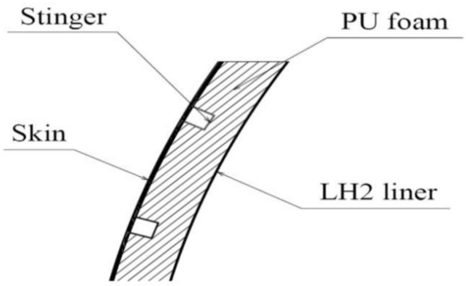


Figure 35: Placement of Insulation and Liners

Aluminium is used as the primary material to compose the storage tank, this stems from the fact that Aluminium shows minimal susceptibility for hydrogen embrittlement. The interior insulation being used is inner wetted thermal insulation spray, a spray on polyurethane foam with embeded metallic liners, specifically designed for this use case. This foam has a thermal conductivity of 0.00675 W/m K, acceptable for our use case.

Lastly, as mentioned before variations in altitude and pressure result in a net energy gain/loss to the storage tanks, along with the heat gained from the catwalk of the storage tanks, the tank storage pressure is subject to fluctuate. To maintain storage pressure, a venting pressure of 25 psi is used. Thus, pressure was selected as it provided the most aerodynamic and cost-effective option, a lower venting pressure would result in the loss of significant amounts of hydrogen fuel while higher venting pressures requires thicker structures.

6. Performance

6.1 Take Off & Landing

6.1.1 Take-Off Distance

The take-off distance consists of two parts, the ground run, and the distance from where the vehicle leaves the ground to until it reaches 50 ft (or 15 m). The sum of these two distances is considered the take-off distance.

The ground run can be calculated using the equations of motion for the vehicle as it moves down the runway. A free-body diagram of the vehicle shows the forces that act on it are the aerodynamic forces of lift and drag (L and D), the thrust force (T), the ground normal force ® and the ground friction force (μR), where μ is the coefficient of rolling friction.

The equations of motion along the runway and perpendicular to it can be written as:

Vertical Distance:

$R - W - L = 0$

Horizontal:

$T - D - \mu R = ma$

Rearranging, we get:

$a = g[(T/W) - \mu] - (g/W)\,(1/2)\,\rho V2S[CD - \mu CL]$

where CL is the lift coefficient during the ground run and CD is the corresponding drag coefficient.

6.1.2 Landing Distance

The landing distance is the distance travelled by the aircraft from the point it touches down to the point it comes to a complete stop. It can be calculated using similar principles as the take-off distance, but in reverse. The aircraft decelerates due to aerodynamic drag, ground friction, and possibly reverse thrust.

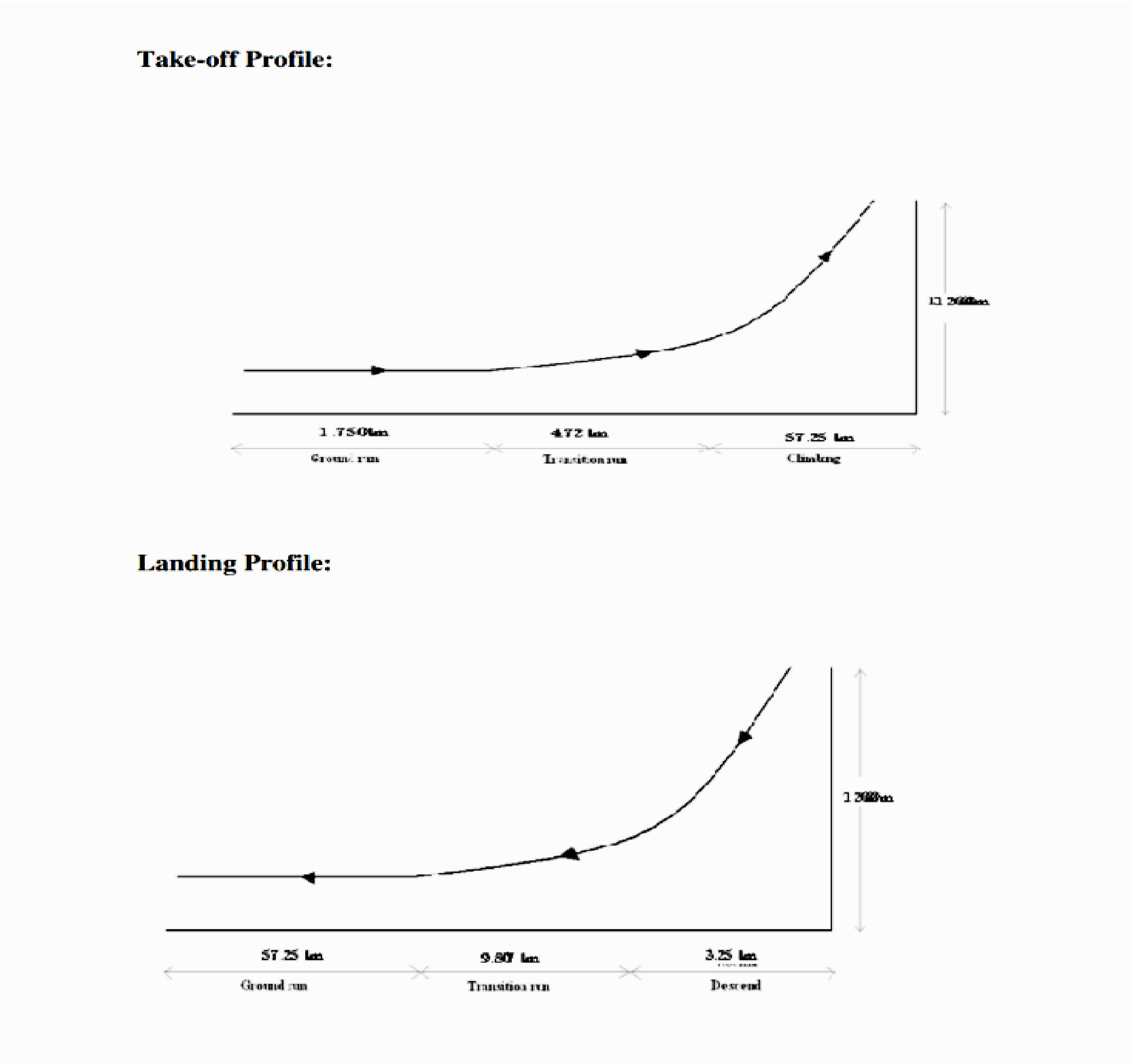


Figure 36: Passenger Aircraft Take-Off and Landing Profile

6.1.3 Ground Run

LENGTH OF TAKE-OFF RUN

a) Ground run (s_1);

$$T - D = \mu (W - L) + \left(\frac{W}{g}\right)\left(\frac{dv}{dt}\right)$$

$$T - D - \mu (W - L) = \left(\frac{W}{g}\right)\left(\frac{dv}{dt}\right)$$

$$s_1 = \int_0^{s_1} ds$$

$$s_1 = \left(\frac{\left(\frac{W}{g}\right)}{T - D - \mu (W - L)}\right)\int_0^{v_1} V\,dv$$

$$s_1 = \left(\frac{W\left(\frac{V_1^2}{2g}\right)}{(T - D) - \mu (W - L)}\right)$$

Figure 37: Formula for Ground run.

$V_1 = 1.2\,V_{stall}$

$V_1 = 1.2 \times 59.16$

$V_1 = 70.992$ m/s

$T = 570.4 \times 10^3$

$D = 525.74 \times 10^3$

So,

$$S_1 = \left[\frac{1.898 \times 10^6 \left(\frac{(70.992)^2}{2 \times 9.18}\right)}{(570.4\times10^3 - 525.74\times10^3) - 0.4\,(1.898\times10^6 - 12.186\times10^6)}\right]$$

$= 1.5341$ Km.

Hence, the minimum distance for take-off run is:- 1.5341 Km ≈ 1.5 Km.

Figure 38: Calculation of Ground run.

The calculated the minimum distance for the Take-Off to happen with maximum payload capacity under normal ISA conditions the Take-Off distance is approximately 1.5Km (Note: This distance will vary depending upon the atmospheric conditions and payload capacity)

6.2 Drag Estimation

A detailed explanation of how various types of drag can be estimated for the Airbus A220 aircraft using design calculations:

1. **Parasitic Drag**: Parasitic drag is the sum of form drag and skin friction drag. Form drag is caused by the shape of the aircraft as it moves through the air. Skin friction drag is caused by the air's friction against the aircraft's surface. These types of drag can be estimated using wind tunnel testing data and computational fluid dynamics (CFD) simulations.
2. **Induced Drag**: Induced drag is caused by the generation of lift. The higher the lift, the higher the induced drag. It can be calculated using the formula: D = 0.5 * Cd * ρ * V^2 * S, where Cd is the drag coefficient, ρ is the air density, V is the velocity, and S is the reference area (wing area). The lift coefficient (C_L) at different flight stages (take-off, cruise, landing) can be used to estimate the induced drag.
3. **Wave Drag**: Wave drag occurs when the aircraft is flying near or above the speed of sound. It can be estimated using the Prandtl-Glauert transformation, which corrects the lift and drag coefficients at high speeds.
4. **Trim Drag**: Trim drag is caused by the use of control surfaces to balance the aircraft. It can be minimized through careful design and can be estimated using stability and control analysis.

For the Airbus A220, we have used the given parameters and the general drag equation to calculate the drag at take-off and landing. The calculated drag at take-off for the Airbus A220 is approximately **63418.99 N**, and at landing is approximately **124518.96 N**. These calculations are based on the given air density, velocity, reference area, and lift coefficient at take-off and landing.

Also, we took help from the Aerodynamic Data reference charts available from the OEM and flight regulation authorities.

220

JET AIRCRAFT

	AIRCRAFT TYPE	1st flight prototype	Aspect ratio A	Taper ratio λ	Sweep angle $\Lambda_{.25}$ deg.	Geom. twist ϵ_g deg.	Dihedral Γ deg.	Profile type and streamwise thickness: root %	Profile type and streamwise thickness: tip %	$(\overline{t/c})$ %	V_{MO}*** km/h EAS	M_{MO}	V_D km/h EAS	M_D	Flap type* T.E./L.E.	$(\overline{c_f/c})$ streamwise	b_f/b %	Flap angle: takeoff deg.	Flap angle: landing deg.	$C_{L_{max}}$ (flight test): takeoff	$C_{L_{max}}$ (flight test): landing
JET TRANSPORT AIRCRAFT	Yakovlev YAK 40	1966	9	.396	0	-	4°30'	15	10	12,5		.70	-	-	P	31.5	67	-	-	-	2.10
	VFW-Fokker 614	1971	7.22	.402	15	3°45'	3°	63_2A-015 a=.4 mod.	65_1A-012 a=.5 mod.	13,5	528	.650	615	.740	F1	31	69	20	40	2.12	2.37
	Grumman Gulfstream II	1966	5.97	.370	25	-	3	13 Naca-6 series		-	680	.860	-	-	F1	30	73	20	40	-	1.80?
	Fokker-VFW F 28	1967	7.27	.355	16	-	2°30'	13.2	10	13	611	.750	-	.830	F1-2	32	69.5	25	42	2.16	2.53
	BAC One-Eleven Srs 200/400	1963	8.00	.321	20	-	2	12.5	11	11,8	652*	.780	760	.860	F1	30	75	18	45	1.86	2.40
	McDonnell-Douglas DC-9 srs 10	1965	8.56	.246	24	-4	3	13.65	9.8	11,65	630	.840	-	.890	S2	36	67	15	50	-	2.40
	McDonnell-Douglas DC-9 srs 30	1966	8.72	.226	24	-4	3	13.65	9.6	11,6	630	.840	-	.890	S3/I	36	67	15	50	2.45	2.98
	Tupolev Tu 134/134A	1964	7.42	.287	35	-	-1°30'	-	-	-	-	.817	-	-	S2	23	60.5	25	30	1.51	1.67
	Boeing 737 srs 100/200	1967	8.83	.251	25	-	6	14	11.5	12,9	648	.840	722	.890	F3/I,II	29	74	15?	40	2.20	3.05
	Aérospatiale Caravelle	1955	8.02	.354	20	-2	3	65_1-212	65_1-212	12	556	.810	-	.870	S1	27	66	10	45	1.74	2.10
	Hawker Siddeley Trident 2E	1967	6.57	.240	35	-5	3	11.5	-	9,8	630	.880	732	.950	S2/I	28	69	20	50	1.93	2.40
	Boeing 727 srs 100/200	1963	7.67	.323	32	-	3	13	9	11	722	.900	-	.950	F3/I,II	30	74	25	40	2.35	2.75
	Tupolev Tu 154	1968	8.14	.250	35	-	-	-	-	-	575	.900	-	-	S3/I	31	75	-	-	2.05	2.70
	Boeing 707/720	1957/60	7.11	.293	35	-	7	12.5	9	10	710*	.900	787	.950	F1/I,II	27	67	15	50	1.76	1.94/2.06
	McDonnell-Douglas DC-8 srs 10,50,61	1958/60/66	7.30	.244	30	-	6°30'	12	10.16	11,1	630	.880	751	.950	S2/II	30	76	25	50	1.87	2.03
	McDonnell-Douglas DC-8 srs 62/63	1966/67	7.65	.194	30	-	6°30'	-	-	-	-	-	770	.950	S2/II	30	73	23	50	1.87	2.05
	BAC VC-10 srs 1100/Super VC-10	1962/64	7.49	.273	32°30'	-	3	12.5	9.75	10	562	.860	704	.940	F1/I	29	62	-	45	-	2.27
	Lockheed L-300 Starlifter	1963	7.90	.350	25	-5°30'	-1°26'	0013 mod.	0010 mod.	11,5	-	-	760*	.890	S2/I	32	66	-	-	-	2.32
	Ilyushin IL 62	1963	6.675	.262	35	-	-	-	-	-	-	-	-	-	S2	24	72	-	30	-	2.36
	BAC Three-Eleven	1972	8.00	.255	25	-	-	12.5	10	14,2	656*	.840	-	-	F1/I	-	75	22	45	-	3.19
	A 300B	1972	8.60	.364	28	-	-	15	-	13,5	668	.840	-	-	F2/I	32.5	82	15	35	2.70	3.00
	Lockheed 1011	1970	7.16	.296	35	-3°40'	7°31'/5°30'	12.4	9	10,7	-	.858	-	.950	S2/I	26	77	22	42	2.46	2.65
	McDonnell-Douglas DC-10 srs 10	1970	6.90	.250	35	-	-	12.5	10	11	-	.880	-	-	S2/I	25?	77.5	25	50	2.34	2.73
	McDonnell-Douglas DC-10 srs 30	1971	7.21	.230	35	-	-	12.2	8.4	11	-	.880	-	-	S2/I	25?	77.5	15	55	2.15	2.98
	Boeing 747/747B	1969/70	6.96	.309	37°30'	-3°30'	7	13.44	8	9,4	695	.920	824	.970	S3/II	30	70	20	30	1.89	2.55
	Lockheed L500 Galaxy	1968	7.75	.256	25	-	-5° 6'	12.5	9.7	11,5	648*	.825	745/726*	.875	F1/I	24	72	30	40	2.21	2.63
JET EXECUTIVE AIRCRAFT	Potez-Air Fouga CM170 Magister	1952	7.42	.400	9°50'	-	0	64-219	64-212	15,5	700	-	740	.820	S1	25	59.5	15	40	-	-
	Aermacchi MB 326	1957	5.26	.600	8°22'	-	2°55'	63A-213.7	63A-212	12,9	806	-	-	.800	S1	25	61	-	-	-	1.70
	Cessna Model 500 Citation	1969	7.45	.390	1°10'	-	-	23014 mod.	23012	13	531*	.700	-	-	S1	25	56	20	-	1.55	1.72
	SNIAS 600 Corvette	1970	7.45	.447	20° 6'	-1°13'	3° 6'	13.65	11.50	12,6	657**	.77	695	.82	S2	31	73	15	40	-	2.40
	Cessna 318 (A/T37)	1954	6.20	.681	≈ 0	-2°38'	3	2418 mod.	2412 mod.	16	816**	.705**	843	-	S1	24	53	-	-	-	1.93
	Gates Lear Jet Model 25	1969	5.02	.510	13°	-	2°30'	64A 109	64A 109	9	617	.765	833	.850	S1	28	61	20	40	1.37	1.37
	Aerocommander Jet Commander	1963	6.19	.333	4°37'	-2	2	64_1-212	64_1-212	12	667	.800	833	.890	P	25	60	20	60	-	2.00
	Dassault Minifalcon 10	1970	7.11	.263	≈27°45'	-	-	-	-	-	-	.87	-	-	S2/I	31.5	67	15	55	2.04	2.35
	MBB 320 Hansa Jet	1964	6.00	.333	-15	-	6	65A-1.5-13	63A-1.8-11	12	825	-	700	.830	S2	22.5	67	-	-	1.61	1.91
	Piaggio Douglas PD 808	1964	6.25	.410	1°50'	-	3	DES0010-1.1-40/11° mod.	DES0008-1.1-40/9° mod.	9	852**		788	.850	S1	30	61	-	-	-	1.71
	North Am. Rockwell Sabre Liner	1958	5.77	.313	28°33'	-	-	-		--	617	.765	-	.850	S1/I			-	-	-	2.37
	Hawker Siddeley HS.125 srs 400B	1962	6.25	.300	20	-2°30'	2	14	11	12,5	528	.755	685	.825	S2	29	65	15	50	1.84	2.19
	Dassault Fan Jet Falcon 20.F	1963	6.40	.312	30	-	2	10.5	8	9,3	685**	.850	-	-	S1/II	25	61	10	40	-	2.32
	Lockheed Jetstar	1957	5.47	.335	30	-2	2	63A 112	63A 309	10,5	710**	.870	787	.90	S2/II	21	69.5	-	-	-	1.67

Table 7-1. Wing design data

*CAS or IAS
**maximum level flight speed
***V_{NE} is taken equal to V_{MO}

*P - plain or split flap
S - slotted flap; 1, 2 or 3: number of slots
F - Fowler flap
I - slat
II - droop nose, leading

Figure 39:Wing-Design Data

Table 14: Lift-Coefficient with Flap Configuration

INCREASE OF LIFT COEFFICIENT WITH FLAPS

Trailing	Leading	α^{TO}	α^{LD}	$C_{L,max}^{TO}$	$C_{L,max}^{LD}$
plain flap		20°	60°	1.60	2.00
single-slotted flap		20°	40°	1.70	2.20
Flower single-slotted flap		15°	40°	2.20	2.90
Flower double-slotted flap		20°	50°	1.95	2.70
Flower double-slotted flap	w/ slat	20°	50°	2.60	3.20
Flower triple-slotted flap	w/ slat	20°	40°	2.70	3.50

Drag Calculation

6.2.1 At Take-Off

Given:

- Air density**, ρ = 1.225 kg/m³** (at sea level)
- Velocity, V = 0.7 x Vlo = 0.7 x 1.2 x Vstall
- Reference area, S = 112.3 m²
- Lift coefficient at take-off, $C_{L(Take\text{-}off)}$ = 2.50 (flaps extended and kept at the take-off position of 20°)

The general drag equation is:

$D=0.5*Cd*\rho*V^2*S$

Substituting the given values into the equation, we get:

$D=0.5*(0.0030+0.2314*(2.50)^2/(3.14*8.6*0.8))*1.225*(0.7*1.2*66.86)^2*112.3$

Solving this equation, we get:

D=63418.99N

So, the calculated drag at take-off for the Airbus A220 is approximately **63418.99 N**.

6.2.2 At Cruise

- ρ = 0.3715 (at the cruising altitude of 10800m)
- V = 242.2 m/s
- S = 400.72 kg/m2
- $C_{L\,(cruise)}$ = 0.63022 (from the wing and aerofoil estimation)

The general drag equation is:

$D=0.5*C_d*\rho*V^2*S$

Substituting the given values into the equation, we get:

D= 0.5∗ (0.0030+0.2314∗ (0.63022)2/ (3.14∗8.6∗0.8)) ∗0.3715∗ $(242.2)^2$∗ 400.72

Solving this equation, we get:

D=72049.275 N

So, the calculated drag at landing for the Airbus A220 is approximately **72049.275 N**.

6.2.3 At Landing

Given:

- Air density, $\rho = 1.225$ kg/m³ (at sea level)
- Velocity, $V = 0.7 \times V_t = 0.7 \times 1.3 \times V_{stall}$
- Reference area, S = 112.3 m²
- Lift coefficient at landing, $CL_{landing} = 3.58$ (flaps extended and kept at the landing position of 40°)

The general drag equation is:

$D=0.5*Cd*\rho*V^2*S$

Substituting the given values into the equation, we get:

D=0.5∗ (0.0030+0.2314∗ (3.58)2/ (3.14∗8.6∗0.8)) ∗1.225∗(0.7∗1.3∗60.55)2∗112.3

Solving this equation, we get:

D=124518.96N

So, the calculated drag at landing for the Airbus A220 is approximately **124518.96 N**.

Note: The Drag-Coefficient does not include the drag caused due to open Landing-Gears as we have calculated the Drag-Coefficient during final descent phase and the time till the Landing Gears are not extended.

8. Conclusion and Future Work

VELOCITY VISTA PROJECT is a short range, high-capacity aircraft that can carry 230 passengers in a two-class layout. Currently, it is also capable of traveling a distance of 3500 nautical miles with maximum payload, with even more increased range in maximized fuel loading conditions. For the propulsive systems, a rear mounted twin engine configuration was chosen with the newly developed General Electric GE 2B67B engine. 19% of an airline's operating cost comes from fuel; therefore, to meet a cost-efficient aircraft an engine with a high bypass and pressure ratio was selected. Furthermore, to reduce the overall carbon emissions and save capital on carbon tax, alternative fuels are also being considered. A trade off study was conducted on the viable use of hydrogen as a propellant, further in Aircraft Design Project II the energy required to transport a fully loaded aircraft and the extra distance travel with the same amount of Hydrogen fuel instead of jet fuel will be calculated. However, in order to make this feasible, certain key technologies need to come into effect. For efficient storage, fuel tanks located in the wings need to be able to withstand the high pressures to ensure that the hydrogen remains in liquid state. Heat exchangers need to be developed with the specialized purpose of heating the hydrogen without causing an explosion. Lastly infrastructure in airports as well as hydrogen production plants need to be scaled, improved, and/or developed to meet the strong demand poised by the airline industry.

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10. Appendix

10.1 **Gross Weight Estimation**

We are going to find the weight estimation of the Aircraft A220, according to the International Standards of FAA.

Gross weight = payload weight + crew weight + empty weight + fuel weight

MTOW (Wo)=W(empty)+W(fuel)+W(payload)+W(additional)

Where:

- *MTOW* = Maximum Take-off Weight
- W_{empty} = Empty Weight of the Aircraft (including structure, systems, and fixed equipment)
- W_{fuel} = Weight of Fuel on Board
- $W_{payload}$ = Weight of Payload (passengers and cargo)
- $W_{additional}$ = Additional Weight (such as catering, crew, baggage, etc.)

Wo= W(crew) + W(payload) + $(WfWo)$Wo + $(WeWo)$Wo

Wo = $Wcrew+Wpayload1-(WfWo)-(WeWo)$

Crew weight = 230

Pay load weight = 40,000 kg (A220)

Fuel weight = 79000 kg (A220)

As our aircraft has sweep angle, K = 1.00 (for Fixed Sweep)

Average of crew weight taking 91 kg/person.

Crew weight = (2+7) *91 = 819 kg = 1805.59 lb

Passenger weight on average of 91kg/person including clothing and baggage = 230*91 =21850kg = 48171.0015 lb

10.2 Empty Weight Estimation

$$W_e/W_0 = A\,W_0^C K_{vs}$$

Where, A= 1.02 C= -0.06

We/Wo = 1.02(661386.78)-0.06 *1 = 0.45

Now,

661386.78 = (1805.59*48171.0015)/ [(1 - (Wf/Wo) - 0.49775)]

661386.78 = 49976.594 / 0.51 - (Wf/Wo)

0.51 - (Wf/Wo) = 49976.594/661386.78

Wf/Wo = 0.417

Now from fuel fraction chart as per IATA and ICAO standards for transport jets:

1. Engine start and warmup – 0.990
2. Taxi – 0.990
3. Take off – 0.995
4. Climb – 0.980
5. Descent – 0.990
6. Landing, Taxi and Shut down – 0.992

Generally, at the end of mission, the fuel tanks are not completely empty, typically (6-8%) allowance is made for reserve and trapped.

Take-off gross weight (Wo) – 1,60,000 pounds

Conceptual *LD* estimation for jet aircraft:

Jet – 0.866 (mach) x $(LD)max$ = 0.866 x 19.25

Maximum speed of the aircraft = 0.86 Mach

Maximum range = 81000 nautical miles = 15001.2 km

Maximum estimated take-off gross weight = 72574.77 Kg = 159999.97 pounds

The basic form of the Breguet range equation is:

R=(vC). Isp. ln($WinitialWfinal$)

Where:

- R = Range of the aircraft (distance traveled)
- v = Velocity of the aircraft (typically in meters per second)
- C = Fuel consumption rate (typically in kilograms per second)
- Isp = Specific impulse of the engine (a measure of engine efficiency, typically in seconds)
- $W_{initial}$ = Initial weight of the aircraft (including fuel)
- W_{final} = Final weight of the aircraft (including fuel)

10.3 **Drag Calculation**

At Take-Off:

Given:

- Air density, **ρ = 1.225 kg/m³** (at sea level)
- Velocity, V = 0.7 x Vlo = 0.7 x 1.2 x Vstall
- Reference area, S = 112.3 m²
- Lift coefficient at take-off, $C_{L(Take\text{-}off)}$ = 2.50 (flaps extended and kept at the take-off position of 20°)

The general drag equation is:

$D=0.5*Cd*\rho*V^2*S$

Substituting the given values into the equation, we get:

$D=0.5*(0.0030+0.2314*(2.50)^2/(3.14*8.6*0.8))*1.225*(0.7*1.2*66.86)^2*112.3$

Solving this equation, we get:

D=63418.99N

So, the calculated drag at take-off for the Airbus A220 is approximately **63418.99 N**.

At Cruise

- ρ = 0.3715 (at the cruising altitude of 10800m)
- V = 242.2 m/s
- S = 400.72 kg/m2
- $C_{L\ (cruise)}$ = 0.63022 (from the wing and aerofoil estimation)

The general drag equation is:

$D=0.5*C_d*\rho*V^2*S$

Substituting the given values into the equation, we get:

$D= 0.5* (0.0030+0.2314* (0.63022)2/ (3.14*8.6*0.8)) *0.3715* (242.2)^2* 400.72$

Solving this equation, we get:

$D=72049.275$ N

So, the calculated drag at landing for the Airbus A220 is approximately **72049.275 N**.

At Landing

Given:

- Air density, $\rho = 1.225$ kg/m³ (at sea level)
- Velocity, $V = 0.7 \times V_t = 0.7 \times 1.3 \times V_{stall}$
- Reference area, $S = 112.3$ m²
- Lift coefficient at landing, $CL_{landing} = 3.58$ (flaps extended and kept at the landing position of 40°)

The general drag equation is:

$D=0.5*Cd*\rho*V^2*S$

Substituting the given values into the equation, we get:

$D=0.5* (0.0030+0.2314* (3.58)2/ (3.14*8.6*0.8)) *1.225*(0.7*1.3*60.55)2*112.3$

Solving this equation, we get:

$D=124518.96$N

So, the calculated drag at landing for the Airbus A220 is approximately **124518.96 N**.